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Boosting cross-architecture adversarial transferability by enhanced deformation attack

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ABSTRACT

Adversarial examples generated on convolutional neural network (CNN) surrogates often transfer less effectively to vision transformer (ViT) targets than to CNN targets, creating a cross-architecture bottleneck for transfer-based black-box attacks. Existing input-transformation attacks diversify gradient estimation, yet the evaluated baselines still exhibit substantial CNN-to-ViT transfer gaps, motivating a deformation strategy that varies both control-point geometry and boundary support. To address this limitation, this paper proposes the Enhanced Deformation Attack (EDA), which estimates attack gradients over stochastically transformed views of the current adversarial image. Each view samples either a full control-point grid with movable boundary points or an interior center grid, and the displaced control points are remapped to a reflection-padded canvas before thin-plate spline (TPS) resampling. Gaussian noise or brightness adjustment provides complementary appearance variation. All transformations are confined to gradient estimation, so the final adversarial example remains in the original coordinate system and satisfies the prescribed ℓ_∞ perturbation budget. On the ImageNet-Compatible dataset, EDA achieves the highest mean attack success rate (ASR) on ViT targets for all four CNN sources, outperforming the strongest source-specific baseline by 5.8, 11.2, 8.7, and 8.0 percentage points under ResNet-18, Inception-v3, Inception-v4, and Inception-ResNet-v2, respectively. The gains also persist on the full ImageNet validation set and ImageNet-V2 across all evaluated source and target groups. Additional evaluations across four perturbation budgets, targeted settings, five random seeds, defended models, and modern robust models further support the stability and scope of the observed transfer gains. The source code is publicly available at <https://github.com/wjc2400136/EDA>.

1. Introduction

Deep neural networks (DNNs) have become core components in modern visual intelligence systems, including image classification (He et al., 2016; Huang et al., 2017), object detection (Lin et al., 2026; Redmon et al., 2016), localization and perception modules (Sun et al., 2025), and safety-critical applications such as autonomous driving and networked control systems (Ding et al., 2025; Geiger et al., 2012; Yan et al., 2025; Zhong et al., 2025). Despite their strong empirical performance, DNNs remain vulnerable to adversarial examples, in which small input perturbations can induce incorrect predictions (Goodfellow et al., 2015). This vulnerability is particularly concerning in security-sensitive scenarios because a malicious perturbation may propagate through an automated decision pipeline and degrade system reliability (Chen et al., 2026).

In black-box attacks, the attacker does not access the target model's internal architecture or parameters. A practical strategy is therefore to

craft adversarial examples on an accessible surrogate model and rely on adversarial transferability to attack an unknown target (Dong et al., 2018). The key difficulty this paper addresses is not merely whether adversarial examples transfer, but why their transferability deteriorates when the surrogate and target models rely on different visual inductive biases. As depicted in Fig. 1, under the RN-18 source setting, CNN-crafted adversarial examples transfer strongly within the CNN family but less effectively to Transformer-based targets. Specifically, the CNN averages exclude the source-identical target, and the exact group-average ASRs are labeled above the bars. Among the baselines, the CNN-ViT gap ranges from 17.7% for SID to 23.2% for L2T, whereas EDA reduces the gap to 12.4 percentage points while requiring 0.260 s per image. The persistent baseline gaps suggest that many transformations still produce perturbations overly adapted to the surrogate model's feature extraction patterns.

One plausible source of this cross-architecture degradation is the difference between the visual processing mechanisms of CNNs and vision

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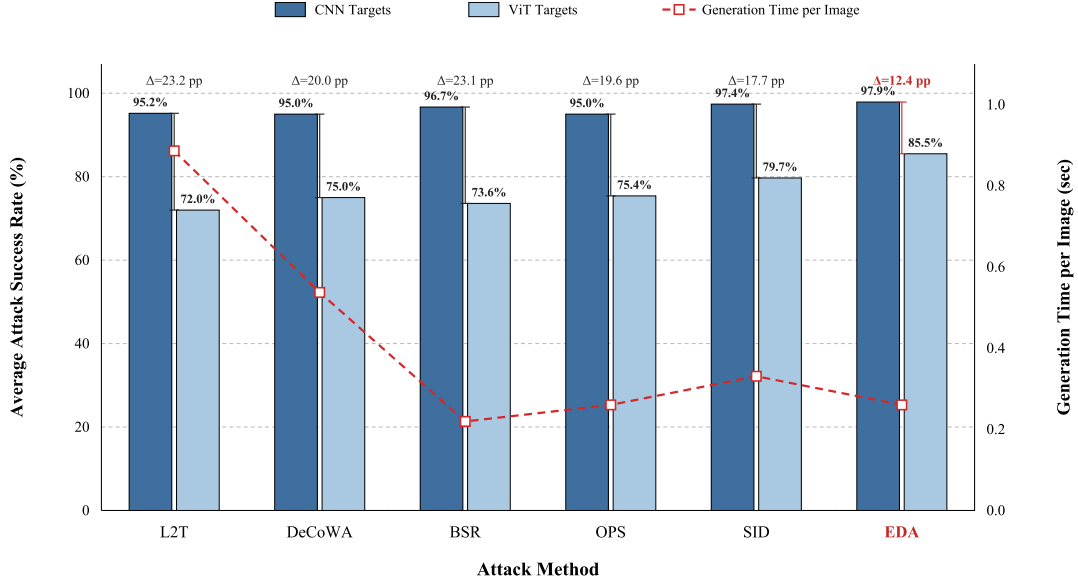


Fig. 1. Cross-architecture transfer imbalance and generation time under the RN-18 source; bar labels give source-excluded group-average ASRs, Δ denotes the CNN-ViT gap in percentage points, and the dashed line gives generation time per image.

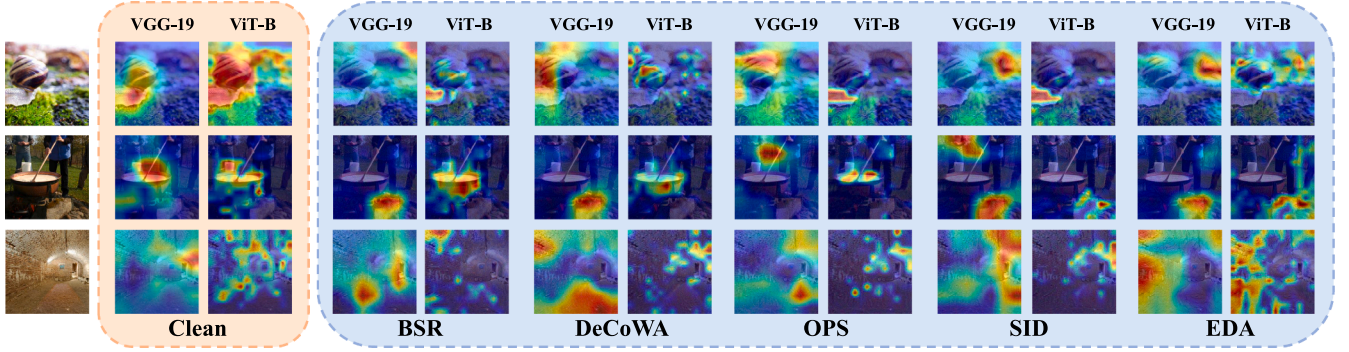


Fig. 2. Grad-CAM maps from VGG-19 and ViT-B for three clean inputs and their adversarial counterparts generated by BSR, DeCoWA, OPS, SID, and EDA; each adjacent column pair corresponds to the two target models.

transformers (ViTs). CNNs aggregate information through local convolutional receptive fields and therefore tend to emphasize local texture and spatially stable patterns. ViTs tokenize images into patches and model long-range relationships through self-attention, making their decisions more dependent on global object layout and inter-region interactions (Dosovitskiy et al., 2021; Geirhos et al., 2019). Fig. 2 provides a qualitative illustration, in which VGG-19 and ViT-B attend to different image regions even on the same clean samples. This visualization is consistent with the architectural-mismatch interpretation but does not establish causality. When adversarial examples are generated using a CNN surrogate, the optimized perturbations can overfit to local texture-sensitive gradients. Such perturbations remain highly effective against CNN targets but may not sufficiently disturb the global reasoning patterns of ViTs.

While both CNNs and ViTs can serve as surrogate models, prior studies have found that adversarial examples crafted on ViTs often transfer more effectively, whereas CNN-crafted adversarial examples remain harder to transfer to ViT targets (Naseer et al., 2022; Wei et al., 2022). This asymmetry motivates this study's focus on the CNN-source setting. Beyond computational efficiency, this setting is also practically relevant: attackers may have access only to lightweight or legacy CNN surrogates deployed on edge devices, embedded cameras, mobile platforms, or local perception modules, while the remote target service may use a larger and undisclosed architecture. Improving CNN-to-ViT transferability is therefore both a technical bottleneck and a realistic threat model.

EDA addresses this failure mode by varying both the spatial support and the control-point geometry used along the gradient path. For each transformed view, the method samples either a full grid with movable boundary points or an interior center grid. It then displaces the selected points and maps them to a reflection-padded canvas whose margins are determined by the deformation scale before thin-plate spline (TPS) resampling. Gaussian noise or brightness adjustment supplies complementary photometric diversity. Because these transformations are used only for gradient estimation, the optimized adversarial example remains in the original coordinate system and satisfies the prescribed ℓ_∞ perturbation constraint.

The main contributions of this paper are summarized as follows:

- Proposing a stochastic TPS deformation strategy that selects one of two control-point layouts with equal probability: a full grid with movable boundary points or an interior center grid.
- Developing a canvas expansion mechanism that determines the width of the reflection padding from the deformation scale and remaps the source and displaced control points before TPS resampling.
- Integrating spatial deformation and appearance augmentation by averaging gradients over independently sampled transformed views, while keeping the final adversarial example in the original coordinate system and within the prescribed perturbation budget.

The rest of the paper is organized as follows. Section 2 reviews prior work on untargeted black-box attacks, Section 3 details EDA, and Section 4 presents and analyzes the experimental results. The last section concludes the paper.

2. Related work

Adversarial examples were first systematically identified and were later explained from the perspective of locally linear behavior in high-dimensional models (Goodfellow et al., 2015; Szegedy et al., 2014). According to the attacker's knowledge, attacks are commonly divided into white-box and black-box settings. White-box attacks assume access to the target architecture and gradients, whereas black-box attacks operate with limited or no internal knowledge of the target model. Transfer-based attacks are especially practical in black-box scenarios because adversarial examples are crafted on a surrogate model and then evaluated on unknown targets. Beyond image classification, transferability has also been studied in security-sensitive vision tasks such as image tampering localization, indicating a broader reliability concern across visual systems (Cao et al., 2024).

Existing transfer-based attacks can be roughly grouped into gradient-based methods (Dong et al., 2018; Kurakin et al., 2017), input transformation-based methods (Wang et al., 2021a; Xie et al., 2019), ensemble-based methods (Liu et al., 2017), and architecture-based methods (Li et al., 2020). Among them, gradient enhancement and input transformation are the most closely related to the present work.

Gradient-based Attacks. Fast Gradient Sign Method (FGSM) added perturbations in the direction that increases the loss (Goodfellow et al., 2015). Iterative FGSM (I-FGSM) extended this idea to multiple update steps, but its transferability can degrade when the perturbation overfits the source model (Kurakin et al., 2017). Momentum Iterative FGSM (MI-FGSM) introduced a momentum term to stabilize update directions and escape poor local optima (Dong et al., 2018). Later variants, including VMI-FGSM and EMI-FGSM, further refined gradient estimation to improve transferability (Wang & He, 2021; Wang et al., 2021b). Other strategies searched for adversarial examples in flatter loss regions (Gan & Wo, 2025; Ge et al., 2023; Qiu et al., 2026), while recent reversible or privacy-oriented attacks explored transferable perturbation generation in black-box settings (Du et al., 2025). EDA builds on the iterative momentum framework, but its main contribution lies in how it samples transformed inputs for gradient estimation.

Input Transformation-based Attacks. Input transformation improves transferability by augmenting inputs during gradient calculation, thereby reducing overfitting to a single surrogate-specific input pattern. This line of work has developed in two main directions. The first direction uses fixed or randomly sampled transformations. DIM randomly resized and padded inputs, while SIM computed gradients over scaled copies (Lin et al., 2020; Xie et al., 2019). Admix mixed images from non-target classes to diversify gradients (Wang et al., 2021a). More recent methods designed increasingly complex transformation spaces: BSR combined block shuffling and rotation, DeCoWA introduced differentiable elastic warping, OPS sampled combinations of transformation operators, and SID used spatial inconsistencies and multi-scale transformations (Guo et al., 2025; Lin et al., 2024; Wang et al., 2024; Zhou et al., 2025). A complementary frequency-domain strategy, MFI, constructed hard or soft mixed-frequency views by exchanging or interpolating high-frequency components between the current input and sampled auxiliary images, thereby diversifying gradient estimation beyond spatial transformations (Qian et al., 2024). The second direction learns transformations adaptively. AITL trained a transformation network, while L2T learned transformation trajectories through differentiable search (Yuan et al., 2022; Zhu et al., 2024). Such methods can be powerful, but they often introduce additional training or computational overhead. EDA belongs to the first direction but differs from existing methods in its deformation design. It combines stochastic sampling between two TPS control-point layouts with scale-dependent boundary support, while treating appear-

ance augmentation as a complementary source of diversity rather than the primary novelty.

3. Proposed method

3.1. Problem definition

Let F denote a pretrained surrogate classifier, and let $x \in [0, 1]^{C \times H \times W}$ be a clean image with ground-truth label y . In the untargeted setting, the adversary seeks an image x^{adv} that increases the classification loss while remaining within an ℓ_∞ ball of radius ϵ around x :

$$\arg \max_{x^{\text{adv}}} \mathcal{L}(F(x^{\text{adv}}), y), \quad \text{s.t.} \quad \|x^{\text{adv}} - x\|_\infty \leq \epsilon, \quad (1)$$

where $\mathcal{L}$ denotes the attack loss. The targeted variant replaces y with the designated target label and minimizes the corresponding loss.

Contemporary transfer attacks commonly build on iterative gradient-based optimization, such as MI-FGSM (Dong et al., 2018). Such methods update the adversarial example along the momentum-accumulated gradient direction that increases the classification loss. The update rule is given by

$$\begin{cases} g_{t+1} = \mu \cdot g_t + \frac{\nabla_x \mathcal{L}(F(x_t^{\text{adv}}), y)}{\|\nabla_x \mathcal{L}(F(x_t^{\text{adv}}), y)\|_1}, \\ x_{t+1}^{\text{adv}} = \text{Clip}_{x, \epsilon}(x_t^{\text{adv}} + \alpha \cdot \text{sign}(g_{t+1})), \end{cases} \quad (2)$$

where g_t is the accumulated gradient, μ is the momentum decay factor, and α is the step size. The operator $\text{Clip}_{x, \epsilon}$ projects the adversarial example into the ℓ_∞ ball centered at x , ensuring $\|x^{\text{adv}} - x\|_\infty \leq \epsilon$.

EDA retains the objective and perturbation constraint in Eq. (1) and modifies only the gradient-estimation step. Fig. 3 summarizes the resulting pipeline. At each iteration i , EDA constructs K independently sampled views of the current adversarial image x_i^{adv} . To construct each view, EDA samples a control-point layout and the corresponding offsets, performs reflection padding, coordinate alignment, TPS resampling, and central cropping, and then applies a sampled appearance transformation. The loss gradients of the K views are averaged before the momentum update. Thus, these transformations affect only gradient estimation; the updated adversarial example remains in the original coordinate system.

3.2. Fine-grained elastic warping

Fine-grained elastic warping uses thin-plate spline interpolation (Bookstein, 1989). For source control points $\mathbf{C} = \{c_i\}_{i=1}^M$ and displaced points $\mathbf{T} = \{t_i\}_{i=1}^M$, TPS constructs a mapping $f = (f_1, f_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that satisfies $f(c_i) = t_i$ while minimizing the summed bending energy

$$\mathcal{E}_{\text{bend}}(f) = \iint \sum_{d=1}^2 \left[\left(\frac{\partial^2 f_d}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 f_d}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 f_d}{\partial y^2} \right)^2 \right] dx dy. \quad (3)$$

This minimum-bending-energy criterion favors smooth non-rigid mappings. This property is useful for gradient estimation in transfer-based attacks because the transformed views should vary spatial structure without introducing discontinuities that dominate the surrogate gradient. The comparison with alternative deformation families is therefore made with respect to this design requirement rather than as a claim of universal superiority. Affine mappings use one global transformation and cannot represent general local non-rigid displacement patterns. Dense elastic fields provide substantially more spatial degrees of freedom, but their smoothness depends on an additional field-smoothing or regularization choice. In contrast, TPS directly interpolates the sparse sampled control-point displacements while selecting, through Eq. (3), the interpolant with minimum bending energy. Thus, for EDA's requirement of sparse control-point sampling with smooth non-rigid deformation, TPS provides a mathematically defined balance between global coherence and local adaptivity; this argument does not establish that TPS is optimal for every possible attack or deformation model.

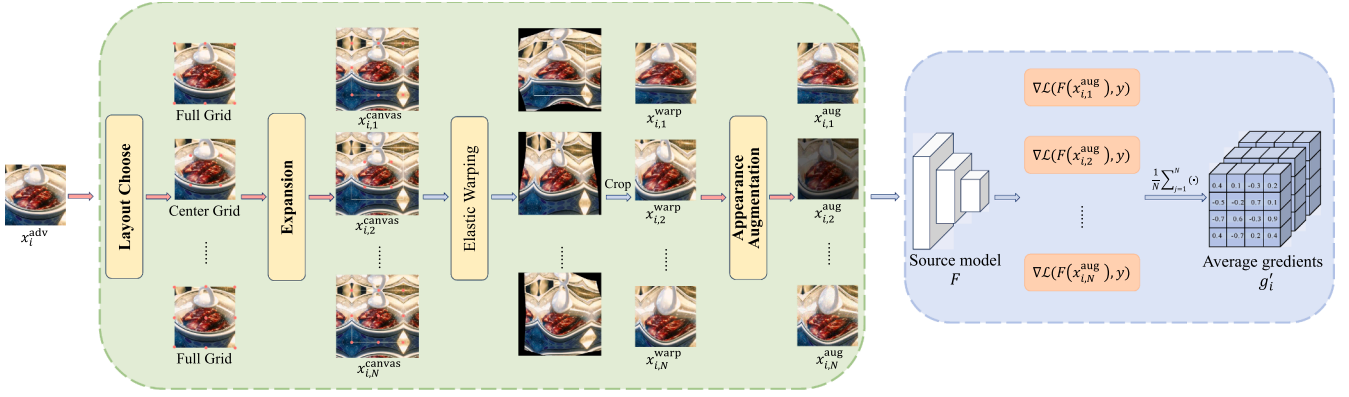


Fig. 3. EDA gradient-estimation pipeline, where each transformed view samples a control-point layout and offsets, applies reflection padding, coordinate remapping, TPS resampling, central cropping, and one appearance transformation, and contributes one gradient to the average used by the momentum update.

For each transformed view, a layout index ℓ determines a source set $\mathbf{C}_\ell = \{c_i\}_{i=1}^{K_\ell}$ in normalized coordinates $[-1, 1]^2$. For each selected point, an offset $n_i = (\Delta x_i, \Delta y_i)$ is sampled independently, where $\Delta x_i, \Delta y_i \sim \mathcal{U}(-s/2, s/2)$. The displaced set is $\mathbf{T}_\ell = \{t_i = c_i + n_i\}_{i=1}^{K_\ell}$. The spatial transformation then applies two coupled operations: layout sampling specifies the control-point geometry, and canvas expansion supplies boundary support before TPS resampling.

3.2.1. Dual control point layout

Let N_h and N_w denote the vertical and horizontal full-grid resolutions. EDA samples one of the following layouts independently for each transformed view, with probability $1/2$ for each branch:

- **Full-grid branch:** $K_{\text{full}} = N_h N_w$ control points are placed at uniformly spaced grid intersections, including the image boundary. Offsets are sampled for every point; therefore, the boundary points move in the default EDA configuration and extend deformation support to boundary regions.
- **Center-grid branch:** $K_{\text{center}} = (N_h - 1)(N_w - 1)$ control points are placed at the cell centers of the full grid. These points lie strictly inside the image domain, so this branch has no boundary control points to fix or move and instead emphasizes interior deformation.

Fig. 4 illustrates the two layouts for $N_h = N_w = 3$.

Boundary-point movement is therefore a property of the full-grid branch, not a third layout or an independent transformation module. When boundary movement is disabled, the implementation sets only the full-grid boundary offsets to zero; the center-grid offsets remain unchanged.

After a branch is selected, offsets are sampled for its K_ℓ points to obtain $\mathbf{T}_\ell$. The tuple $(\ell, \mathbf{C}_\ell, \mathbf{T}_\ell)$ is then passed to the canvas-expansion and coordinate-remapping stage. Alternating between the boundary-inclusive full grid and the interior center grid reduces dependence on a single control-point arrangement.

3.2.2. Adaptive canvas expansion

After layout and offset sampling, the canvas expansion computes the required padding from the deformation scale s and constructs a reflection-padded canvas. Both control-point sets are then expressed in the expanded coordinate system. TPS resampling and central cropping complete the same spatial routine and return an $H \times W$ transformed view x_{warp} for appearance augmentation.

The movable boundary points in the full-grid branch can send the TPS sampling map outside the original image support. Without additional support, such locations are filled by the resampler's padding rule and can introduce black borders or abrupt edge responses. EDA therefore pads the current adversarial image by reflection before constructing the TPS grid and crops the central region after resampling. Fig. 5

contrasts this procedure with warping on the unexpanded canvas. The displayed warps are intermediate inputs used for gradient estimation, not final adversarial examples.

The canvas expansion is derived from the maximum control-point displacement allowed by the deformation scale s . Since control-point coordinates are normalized to $[-1, 1]$, a normalized horizontal displacement Δx corresponds to a pixel displacement of $\Delta x(W - 1)/2$, and a normalized vertical displacement Δy corresponds to $\Delta y(H - 1)/2$. Each component of the control-point offset is sampled from $\mathcal{U}(-s/2, s/2)$, so the largest possible normalized displacement along either axis is $\Delta_{\text{max}} = s/2$. The implementation therefore sets the padding amounts to

$$\begin{cases} p_w = \left\lceil \frac{\Delta_{\text{max}} \times (W - 1)}{2} \right\rceil, \\ p_h = \left\lceil \frac{\Delta_{\text{max}} \times (H - 1)}{2} \right\rceil, \end{cases} \quad (4)$$

where W and H denote the original image width and height. The ceiling operation gives integer-valued padding.

Reflection padding produces $x_{\text{canvas}} \in \mathbb{R}^{C \times H' \times W'}$, where $H' = H + 2p_h$ and $W' = W + 2p_w$. Compared with zero or constant padding, reflection padding preserves local intensity continuation at the canvas boundary and reduces abrupt padding-induced gradients. This padding rule bounds the sampled control-point offsets along each axis; it does not bound every interpolated TPS location because a continuous TPS map may overshoot its control points. The expansion is therefore a boundary-support rule rather than a formal guarantee of artifact-free or semantics-preserving deformation. Because the expanded canvas exists only inside the transformed-loss computation, it does not change the final ℓ_∞ -bounded adversarial output.

Canvas expansion changes both the image dimensions and the normalized coordinate system used by the TPS solver. The selected source and displaced point sets, $\mathbf{C}_\ell = \{c_i\}_{i=1}^{K_\ell}$ and $\mathbf{T}_\ell = \{t_i\}_{i=1}^{K_\ell}$, are therefore converted to the expanded-canvas system before the TPS coefficients are computed. Fig. 6 summarizes the three conversions below.

Here, valid zero-based pixel-center indices range from 0 to $W - 1$ horizontally and from 0 to $H - 1$ vertically.

- **Original-image coordinates.** The endpoint-aligned coordinates of a normalized point $q = (q_x, q_y)$ are

$$\Pi_{W,H}(q) = \left((q_x + 1) \frac{W - 1}{2}, (q_y + 1) \frac{H - 1}{2} \right). \quad (5)$$

This conversion is applied to both c_i and t_i .

- **Expanded-canvas translation.** The source and displaced control points are shifted by the padding amounts:

$$\begin{cases} \tilde{c}_i = \Pi_{W,H}(c_i) + (p_w, p_h), \\ \tilde{t}_i = \Pi_{W,H}(t_i) + (p_w, p_h). \end{cases} \quad (6)$$

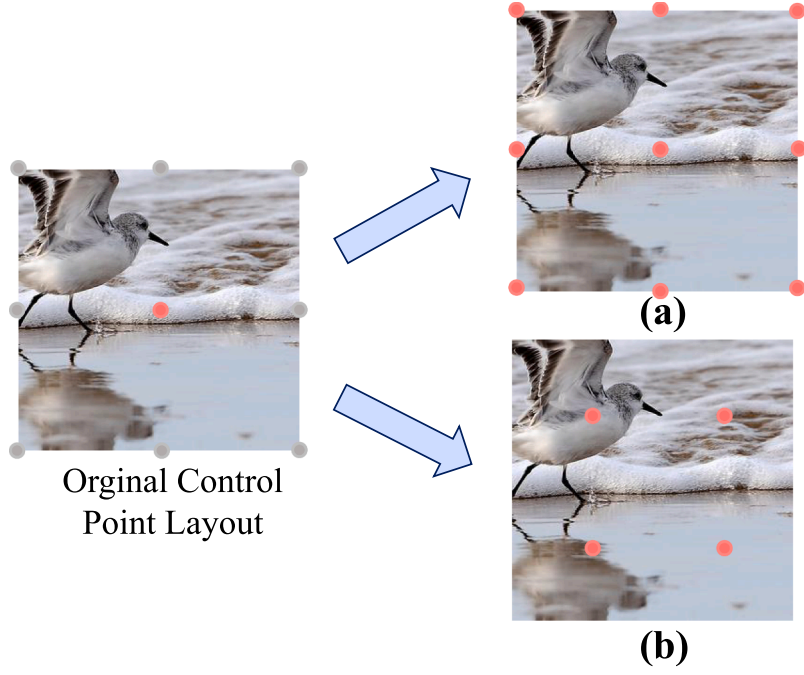


Fig. 4. Example dual control-point layouts for $N_h = N_w = 3$: (a) a 3×3 full grid whose boundary points are movable and (b) a 2×2 center grid located at the full-grid cell centers.

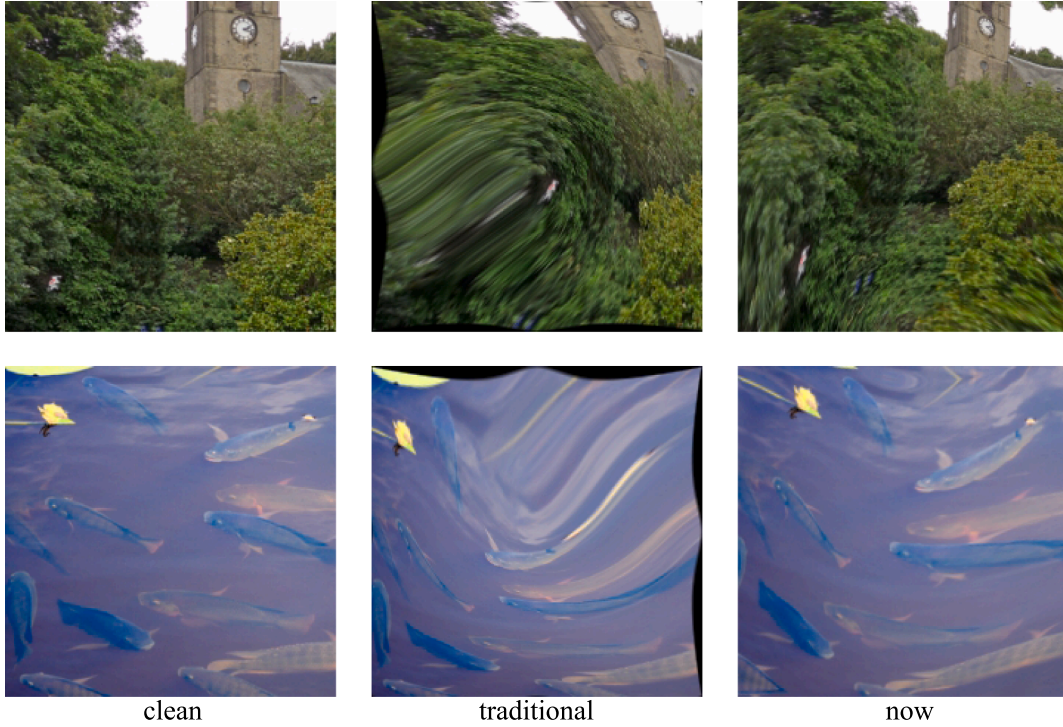


Fig. 5. Boundary effects of TPS deformation on two inputs: clean images (left), warps on the unexpanded canvas (middle), and warps produced with reflection expansion and central cropping (right).

- **Expanded-canvas normalization.** The shifted coordinates $\tilde{c}_i$ and $\tilde{t}_i$ are mapped to the endpoint-aligned normalized domain of the expanded canvas:

$$\begin{cases} c'_i = (2\tilde{c}_{i,x}/(W' - 1) - 1, 2\tilde{c}_{i,y}/(H' - 1) - 1), \\ t'_i = (2\tilde{t}_{i,x}/(W' - 1) - 1, 2\tilde{t}_{i,y}/(H' - 1) - 1). \end{cases} \quad (7)$$

Eqs. (5)–(7) reproduce the coordinate conversions in the implementation. With the aligned sets $\mathbf{C}'_\ell = \{c'_i\}_{i=1}^{K_\ell}$ and $\mathbf{T}'_\ell = \{t'_i\}_{i=1}^{K_\ell}$, the

TPS solver constructs f such that $f(c'_i) = t'_i$. Evaluating f on the dense output grid yields $\mathcal{G} \in \mathbb{R}^{H' \times W' \times 2}$, which is used as the backward sampling grid. Thus, for each output location z , $f(z)$ specifies the coordinate sampled from x_{canvas} . The expanded canvas is then bilinearly resampled and centrally cropped to obtain x_{warp} . In the implementation, resampling uses `align_corners=False`; therefore, the endpoint-based control-point conversion and the resampling convention are reported separately rather than assumed to be pixel-center equivalent.

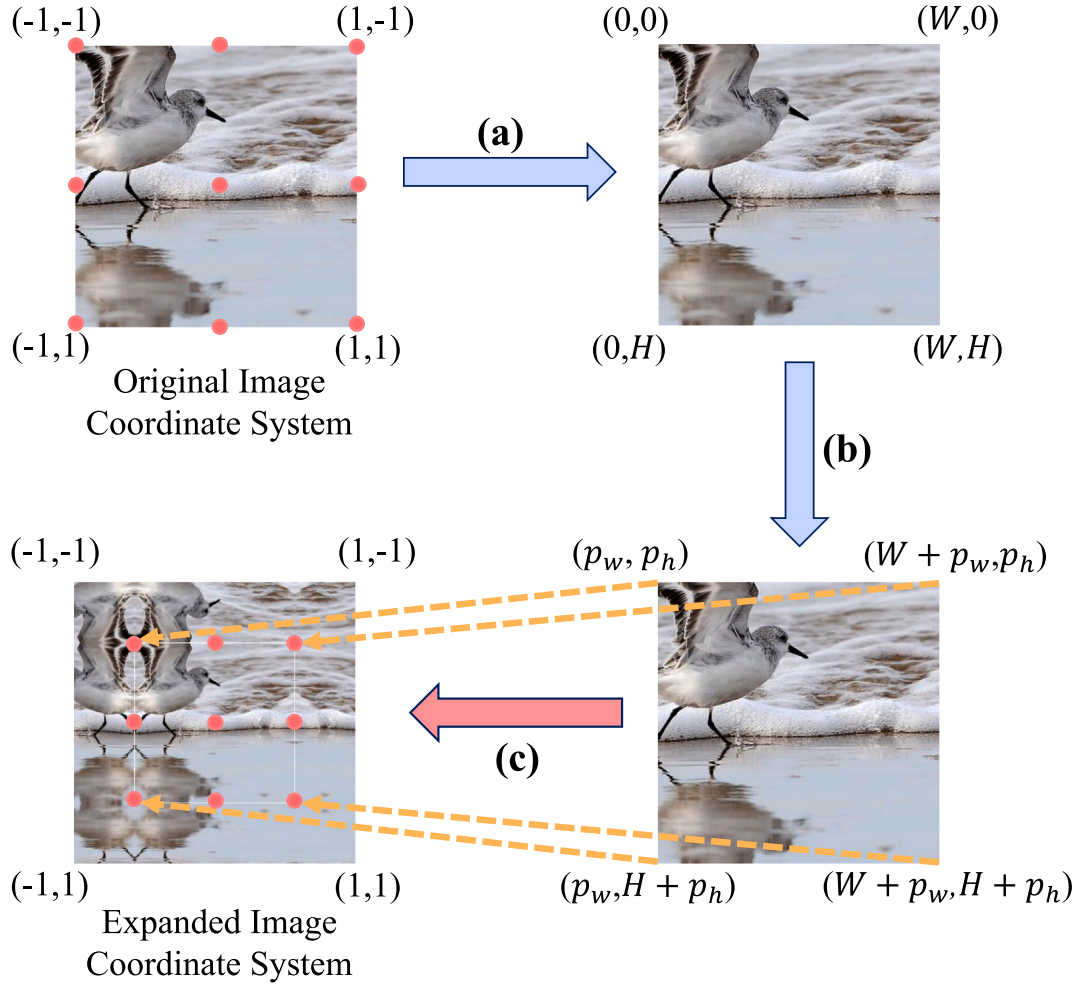


Fig. 6. Control-point alignment between the original and expanded canvases: (a) conversion from normalized coordinates to original-image pixel coordinates, (b) translation by (p_w, p_h) , and (c) conversion to normalized expanded-canvas coordinates.

Operationally, layout selection, offset sampling, reflection padding, coordinate remapping, dense TPS-grid construction, bilinear resampling, and central cropping are executed consecutively within one spatial transformation, denoted by ElasticWarp below. This ordering, which matches the implementation, is represented directly in Algorithm 1.

In Algorithm 1, each pair (i, j) denotes the j th transformed sample at the i th attack iteration. The layout $\ell_{i,j}$, TPS control-point offsets, and appearance operator are sampled before the corresponding forward pass. The operation ElasticWarp comprises reflection padding according to Eq. (4), coordinate alignment, TPS sampling-grid construction, bilinear resampling, and central cropping. These random choices define a transformation path and are held fixed when computing the gradient with respect to the current adversarial image. EDA does not differentiate through the sampling decisions or optimize the sampled transformation parameters. Instead, it estimates the gradient of an expected transformed loss by Monte Carlo averaging:

$$\bar{g}_i \approx \frac{1}{K} \sum_{j=1}^K \nabla_{x_i^{\text{adv}}} \mathcal{L}(F(\tau_{i,j}(x_i^{\text{adv}})), y), \quad (8)$$

where $\tau_{i,j}$ denotes the sampled spatial and appearance transformation. This expression is a Monte Carlo estimator of the expected transformed-loss gradient, not a derivative through the discrete sampling distribution. Conditioned on $\tau_{i,j}$, reflection padding and bilinear TPS resampling allow gradients to propagate to the input image, while additive noise and brightness adjustment are differentiable before clipping. The clipping operation is piecewise differentiable and is handled by the standard

subgradient used in automatic differentiation. Thus, gradients are back-propagated through the sampled transformation path to x_i^{adv} , while the stochastic sampling process itself is not differentiated. The implementation normalizes momentum by the per-sample mean absolute gradient. For a fixed input dimensionality, this differs from the ℓ_1 normalization in Eq. (2) only by a positive constant and therefore leaves the sign-based update unchanged. Algorithm 1 presents the untargeted case; a targeted attack uses the designated target label and reverses the loss sign.

3.3. Appearance augmentation

Appearance augmentation follows the cropped spatial view x_{warp} and returns x_{aug} after applying one randomly selected photometric operator. While fine-grained deformation diversifies spatial structures, this component reduces dependence on a single photometric condition. The design also aligns with recent observations that semantic-preserving appearance changes can expose additional vulnerabilities in intelligent vision systems (Du et al., 2026). Two lightweight operators are considered: pixel-level Gaussian noise and image-level brightness adjustment.

- **Noise Injection (Pixel-level):** Independent Gaussian noise is added to each RGB channel:

$$x_{\text{noise}} = \text{Clip}_{[0,1]}(x_{\text{warp}} + \eta), \quad \eta \sim \mathcal{N}(0, \sigma^2).$$

The standard deviation σ controls the strength of the pixel-level variation.

Algorithm 1: Enhanced deformation attack.

Input: Surrogate model F , clean image x , ground-truth label y , iteration number T , perturbation budget ϵ , decay factor μ , full-grid resolutions N_h and N_w , deformation scale s , number of transformed samples K

Output: An adversarial example x^{adv}

```

1 Initialize:  $x_0^{\text{adv}} = x$ ,  $m_0 = 0$ ,  $\alpha = \frac{\epsilon}{T}$ ;
2 for  $i = 0$  to  $T - 1$  do
3   for  $j = 1$  to  $K$  do
4     Layout Selection:
        $\ell_{i,j} \sim \text{Uniform}\{\text{Full Grid, Center Grid}\}$ ;
5     Offset Sampling: each component of  $\mathbf{n}_{i,j}$  is sampled
       from  $\mathcal{U}(-s/2, s/2)$ ;  $\mathbf{T}_{i,j} = \mathbf{C}_{\ell_{i,j}} + \mathbf{n}_{i,j}$ ;
6     Fine-Grained Elastic Warping:
        $x_{i,j}^{\text{warp}} = \text{ElasticWarp}(x_i^{\text{adv}}, \mathbf{C}_{\ell_{i,j}}, \mathbf{T}_{i,j}, s)$ ;
7     Appearance Selection:  $a_{i,j} \sim \{\text{Noise, Brightness}\}$ ;
8     Appearance Augmentation:  $x_{i,j}^{\text{aug}} = a_{i,j}(x_{i,j}^{\text{warp}})$ ;
9     Compute gradient:  $g_{i,j} = \nabla_{x_i^{\text{adv}}} \mathcal{L}(F(x_{i,j}^{\text{aug}}), y)$ ;
10  end
11  Calculate average gradient:  $\bar{g}_i = \frac{1}{K} \sum_{j=1}^K g_{i,j}$ ;
12  Update momentum:  $m_{i+1} = \mu \cdot m_i + \frac{\bar{g}_i}{\text{mean}(|\bar{g}_i|)}$ ;
13  Update adversarial example:
        $x_{i+1}^{\text{adv}} = \text{Clip}_{x,\epsilon}(x_i^{\text{adv}} + \alpha \cdot \text{sign}(m_{i+1}))$ ;
14 end
15 return  $x^{\text{adv}} = x_T^{\text{adv}}$ ;

```

- **Brightness Adjustment (Image-level):** The warped image is multiplied by a random brightness factor:

$$x_{\text{bright}} = \text{Clip}_{[0,1]}(r \cdot x_{\text{warp}}), \quad r \sim \mathcal{U}(r_{\min}, r_{\max}).$$

The interval $[r_{\min}, r_{\max}]$ controls the range of image-level darkening and brightening and is chosen to include the identity value $r = 1$. Clipping keeps the transformed image within the valid input range.

For each transformed view, one of the two appearance transformations is selected uniformly at random and applied to the warped image:

$$x_{\text{aug}} = \text{Aug}(x_{\text{warp}}), \quad \text{Aug} \sim \{\text{Noise, Brightness}\}.$$

The two augmentations are applied only to intermediate inputs for gradient estimation and do not alter the final perturbation. They complement the proposed structural deformation mechanism and provide additional photometric diversity during gradient estimation.

3.4. EDA vs. DeCoWA

Both EDA and DeCoWA (Lin et al., 2024) use TPS warping to improve adversarial transferability, but their mechanisms differ in four respects.

- **Deformation Paradigm.** DeCoWA treats the 2D control-point offsets $\{n_i\}$ as learnable parameters and optimizes them by gradient descent before updating the adversarial perturbation. EDA instead samples offsets $n_i = (\Delta x_i, \Delta y_i)$ with independent components from $\mathcal{U}(-s/2, s/2)$ and uses the resulting warp as a stochastic input transformation. The adversarial perturbation δ remains the only optimized variable.
- **Spatial Coverage.** EDA introduces two control-point layouts: a full grid and an interior center grid. The full-grid branch includes movable boundary points, whereas the center-grid branch contains only interior points. This exposes the surrogate to deformation patterns defined by both boundary-inclusive and interior control points, reducing dependence on one fixed layout.

- **Boundary Handling.** Scale-dependent reflection expansion supports the movable boundary points of the full-grid branch. The expanded canvas reduces artifacts caused by out-of-support sampling and preserves stable gradient propagation near image boundaries.
- **Algorithmic Flow.** DeCoWA uses a hierarchical optimization process and incurs additional backpropagation for learning deformation parameters. EDA applies pre-sampled deformations directly to the perturbed image and averages the resulting gradients, reducing per-iteration transformation-optimization overhead.

Mechanistically, CNN surrogates can overfit to locally aligned spatial perturbations because their receptive fields emphasize local texture and neighborhood consistency. ViTs, however, aggregate long-range patch interactions through self-attention. EDA addresses this mismatch by optimizing perturbations under a distribution of smooth structural transformations. Sampling between the two control-point layouts prevents the perturbation from relying on one fixed geometry, and scale-dependent canvas expansion supports deformation near the image boundary while reducing artificial border gradients. Appearance augmentation further increases input diversity, but it complements the deformation design rather than replacing it.

4. Experiments

4.1. Experimental settings

Datasets: Three ImageNet-style datasets are used. The main benchmark is the NIPS 2017 adversarial competition’s ImageNet-Compatible Dataset (IN-C-1K), which contains 1000 color images with a ground-truth class label and a designated target class label for each image (Wang et al., 2024). To test whether the results extend beyond this 1K benchmark, evaluation is also performed on the full 50,000-image ImageNet validation set (IN-Val-50K) and the 10,000-image ImageNet-V2 matched-frequency set (IN-V2-10K), an independently collected distribution-shifted dataset. Unless otherwise stated, the main comparisons, ablations, targeted attacks, and multi-seed experiments use IN-C-1K. Untargeted evaluation uses ground-truth labels on all three datasets, whereas the designated target labels supplied with IN-C-1K are used for targeted evaluation.

Models: Four source architectures are used to generate adversarial examples: ResNet-18 (RN-18) (He et al., 2016), Inception-v3 (Inc-v3) (Szegedy et al., 2016), Inception-v4 (Inc-v4) (Szegedy et al., 2017), and Inception-ResNet-v2 (IncRes-v2) (Szegedy et al., 2017). Transferability is evaluated on three target-model groups. The CNN group contains VGG-19 (Simonyan & Zisserman, 2015), RN-18, ResNet-50 (RN-50), ResNet-101 (RN-101) (He et al., 2016), ResNeXt-50 (RNx-50) (Xie et al., 2017), DenseNet-121 (DN-121) (Huang et al., 2017), MobileNet-v2 (MB-v2) (Sandler et al., 2018), Inc-v3, Inc-v4, and IncRes-v2. The Transformer group contains ViT-Base (ViT-B) (Dosovitskiy et al., 2021), DeiT-Base (DeiT-B) (Touvron et al., 2021a), LeViT-256 (Graham et al., 2021), PiT-Base (PiT-B) (Heo et al., 2021), CaiT-Small (CaiT-S) (Touvron et al., 2021b), ConViT-Base (ConViT-B) (D’Ascoli et al., 2021), TNT-Small (TNT-S) (Han et al., 2021), Visformer-Small (Vis-S) (Chen et al., 2021), Swin-Tiny (Swin-T) (Liu et al., 2021), and CoaT-Tiny (CoaT-T) (Xu et al., 2021). The defense-oriented group contains ten targets. It includes two adversarially trained RN-50 checkpoints: one trained under an ℓ_2 perturbation constraint ($\|\delta\|_2 \leq 1$) and the other under an ℓ_∞ perturbation constraint ($\|\delta\|_\infty \leq \frac{4}{255}$). The group also includes FAT under 2-pixel and 4-pixel settings (Tramèr et al., 2018). The remaining targets are the normally trained Inc-v3 and IncRes-v2 checkpoints from the timm repository (Wightman, 2019), HGD (Liao et al., 2018), NRP with RN-101 and ViT-B backbones (Naseer et al., 2020), and randomized smoothing (RS) (Cohen et al., 2019). To distinguish the two normally trained checkpoints from adversarially trained models, the suffix “(timm)” is used throughout the defense evaluation.

Table 1

Attack success rates (%) transferred from four CNN sources to ten CNN targets on the ImageNet-Compatible dataset; source-identical entries are marked with an asterisk and excluded from the corresponding averages.

Source	Attack	Target model										Avg.
		VGG-19	RN-18	RN-50	RN-101	RNX-50	DN-121	MB-v2	Inc-v3	Inc-v4	IncRes-v2	
RN-18	L2T	<u>99.3</u>	100.0*	94.4	92.0	92.8	99.5	98.3	95.0	94.5	91.1	95.2
	BSR	<u>99.3</u>	100.0*	96.2	<u>94.8</u>	95.1	99.6	<u>98.8</u>	95.5	96.7	94.0	96.7
	DeCoWA	98.8	100.0*	93.1	91.2	92.1	99.3	97.7	94.5	95.3	93.1	95.0
	OPS	98.7	100.0*	92.5	90.8	91.7	99.3	97.5	95.9	95.2	93.6	95.0
	SID	99.1	100.0*	<u>96.9</u>	<u>94.8</u>	<u>95.6</u>	99.8	98.6	<u>97.1</u>	98.0	96.4	<u>97.4</u>
	EDA	99.7	100.0*	97.0	96.9	97.0	<u>99.7</u>	99.2	97.7	<u>97.7</u>	<u>96.3</u>	<u>97.9</u>
Inc-v3	L2T	94.4	<u>93.4</u>	86.7	84.4	<u>86.7</u>	95.6	89.8	99.8*	95.9	94.5	<u>91.3</u>
	BSR	92.6	92.7	83.1	79.3	83.3	94.5	88.2	100.0*	95.3	92.6	89.1
	DeCoWA	91.5	91.5	78.5	77.5	77.8	94.9	85.3	99.8*	95.8	92.2	87.2
	OPS	91.5	91.8	78.8	77.9	80.2	94.7	85.9	99.8*	96.0	94.0	87.9
	SID	93.8	93.1	82.5	81.8	84.8	<u>96.0</u>	88.9	100.0*	<u>96.8</u>	<u>95.2</u>	90.3
	EDA	95.1	95.2	91.7	91.2	91.7	97.9	92.7	<u>99.9*</u>	98.4	97.5	94.6
Inc-v4	L2T	90.5	88.4	75.7	74.3	78.6	90.7	87.0	92.2	99.2*	91.3	85.4
	BSR	92.1	87.3	74.3	71.6	75.8	91.4	85.8	90.7	100.0*	88.7	84.2
	DeCoWA	84.2	83.2	63.5	58.8	64.2	84.7	79.1	86.8	98.7*	86.6	76.8
	OPS	89.1	<u>90.2</u>	73.5	74.7	77.5	90.9	85.4	93.2	98.3*	91.3	85.1
	SID	93.4	92.3	80.5	80.9	83.7	95.4	<u>90.2</u>	<u>94.9</u>	<u>99.9*</u>	<u>95.1</u>	89.6
	EDA	<u>93.2</u>	92.3	85.7	86.9	88.3	<u>95.0</u>	91.2	96.2	99.6*	95.8	91.6
IncRes-v2	L2T	88.9	88.2	77.9	75.8	77.2	89.5	86.3	91.5	93.1	98.8*	85.4
	BSR	92.3	90.6	77.9	75.5	79.0	93.1	88.9	94.6	96.3	99.7*	87.6
	DeCoWA	82.9	83.7	63.0	60.1	63.5	85.1	80.0	87.2	89.5	97.2*	77.2
	OPS	86.3	87.4	73.0	73.5	75.1	89.4	83.9	91.4	91.4	96.7*	83.5
	SID	<u>91.8</u>	92.4	81.6	81.5	<u>83.4</u>	<u>94.6</u>	89.7	<u>94.7</u>	96.1	<u>99.4*</u>	<u>89.5</u>
	EDA	91.5	<u>91.1</u>	88.2	87.3	87.6	95.2	<u>89.6</u>	95.8	<u>96.2</u>	<u>99.4*</u>	91.4

Attack Settings: Evaluation is conducted against five representative recent methods: L2T (Zhu et al., 2024), BSR (Wang et al., 2024), DeCoWA (Lin et al., 2024), OPS (Guo et al., 2025), and SID (Zhou et al., 2025). L2T retains its original configuration. In OPS notation, $\text{OPS}(T, N_p, N_e)$ uses T attack iterations, N_p operator samples, and N_e perturbation samples per iteration, for $T(N_p N_e + 1)$ gradient computations. To align the transformation-sampling scale approximately with $\text{OPS}(10, 5, 5)$, BSR, DeCoWA, SID, and EDA each use $K = 25$ transformed views per iteration; all remaining method-specific hyperparameters follow their published settings.

EDA Settings: The full-grid resolution is $N_h = N_w = 3$, giving nine full-grid points and four center-grid points. The two layouts are sampled with equal probability. All full-grid boundary points are movable in the default method. The deformation scale is $s = 0.45$, Gaussian appearance noise uses $\sigma = 0.05$, and the brightness factor is sampled from $\mathcal{U}(0, 2)$. Random-start initialization is disabled. At attack iteration i , the stochastic transformations are sampled using seed $b + i$, where b denotes the base seed. The main single-run experiments use $b = 42$, while the stability study uses $b \in \{0, 1, 2, 3, 4\}$.

General Settings: Adversarial examples are generated with the MIFGSM update using maximum perturbation $\epsilon = \frac{16}{255}$, $T = 10$ iterations, step size $\alpha = \frac{\epsilon}{T} = \frac{1.6}{255}$, and momentum decay factor $\mu = 1$. Cross-entropy is used as the attack loss. Pixel values are normalized to $[0, 1]$; therefore, both ϵ and α are reported in normalized pixel units. Unless otherwise specified, the single-run experiments use random seed 42. Attack performance is quantified by attack success rate (ASR). All experiments were conducted on one NVIDIA RTX 3090 GPU.

Averaging Protocol: Unless otherwise specified, when an average ASR is used to measure transferability over a target group, any target model identical to the source model is excluded. The corresponding self-transfer result is still reported when available and marked with an asterisk. For defended or robustness-oriented models, a target is excluded only when it uses the same checkpoint as the source; defended, re-trained, or preprocessing-wrapped models with different parameters are treated as distinct transfer targets. When an overall Avg. is computed from the CNN and ViT group averages, the two group averages are

equally weighted. Reported averages and improvements are computed from unrounded values unless otherwise stated.

4.2. Attacks on the ImageNet-compatible dataset

To establish the main transfer benchmark, EDA is compared with five representative baseline attacks using four ImageNet-pretrained CNN surrogate models. The evaluation first considers transfer within the CNN family and then the central CNN-to-ViT setting. Within each comparison, the best result is boldfaced, and the second-best result is underlined.

4.2.1. Using CNNs to attack CNNs

To assess whether EDA improves transferability without changing the architectural family, the six attacks are evaluated against ten CNN targets. The results in Table 1 demonstrate that EDA obtains the highest source-excluded average ASR for all four CNN sources, reaching 97.9%, 94.6%, 91.6%, and 91.4% for RN-18, Inc-v3, Inc-v4, and IncRes-v2, respectively. The RN-18 setting is close to saturation for several attacks, so the average margin is relatively small: EDA reaches 97.9%, compared with 97.4% for SID. More challenging differences appear when Inception-family sources transfer to residual-family targets. With Inc-v4 as the source, EDA obtains 85.7%, 86.9%, and 88.3% on RN-50, RN-101, and RN-50, but 96.2% and 95.8% on Inc-v3 and IncRes-v2. A similar pattern appears under the IncRes-v2 source. These target-wise gaps indicate that source-target architectural affinity still affects transfer even within CNNs. The lower ASRs should not be interpreted as intrinsic robustness of the residual-family targets: the pattern is source-dependent, because the Inception-family sources transfer less effectively to RN-50, RN-101, and RN-50 than to the Inception-family targets, whereas the RN-18 source does not exhibit the same ordering. A plausible explanation is weaker alignment of transferable source directions across these architectural families. EDA's diverse smooth deformations can reduce dependence on one exact surrogate-specific spatial alignment and improve the group average, but Table 1 does not isolate this architectural explanation causally or show that the remaining mismatch is eliminated.

Table 2

Attack success rates (%) transferred from four CNN sources to ten ViT-family targets on the ImageNet-Compatible dataset.

Source	Attack	Target model										Avg.
		ViT-B	DeiT-B	LeViT	PiT-B	CaiT-S	ConViT-B	TNT-S	Vis-S	Swin-T	CoaT-T	
RN-18	L2T	51.2	69.9	83.7	61.8	70.3	60.3	74.5	82.0	79.1	87.2	72.0
	BSR	57.0	72.4	87.5	63.6	70.7	60.8	74.3	85.5	77.5	86.5	73.6
	DeCoWA	61.2	74.8	87.4	65.0	74.0	64.9	76.2	84.2	77.1	84.9	75.0
	OPS	58.1	75.1	86.5	65.7	74.7	66.6	77.7	83.6	79.3	86.5	75.4
	SID	64.7	80.7	91.1	67.3	79.6	71.1	82.1	89.5	82.4	88.8	79.7
	EDA	71.9	86.3	92.6	78.7	86.9	77.7	86.9	91.4	89.2	93.8	85.5
Inc-v3	L2T	45.9	62.6	78.0	58.8	65.3	54.5	67.4	74.6	68.2	80.4	65.6
	BSR	42.8	55.4	72.8	51.4	59.8	49.0	60.6	68.1	60.9	74.2	59.5
	DeCoWA	43.1	56.7	71.3	51.1	58.6	49.9	64.1	68.6	60.6	71.0	59.5
	OPS	44.5	56.9	72.5	51.3	59.7	52.4	61.0	66.9	60.9	71.3	59.7
	SID	47.1	60.5	74.3	53.6	62.4	54.7	64.1	71.4	62.5	73.3	62.4
	EDA	60.1	75.7	86.1	70.9	77.7	69.5	78.6	86.5	77.8	85.5	76.8
Inc-v4	L2T	38.3	51.5	67.3	46.9	53.8	43.9	57.1	67.0	58.4	71.3	55.5
	BSR	33.2	47.6	66.2	45.2	50.6	41.0	51.3	61.0	55.0	68.8	52.0
	DeCoWA	30.1	38.8	55.3	34.1	40.4	35.2	45.1	51.3	44.0	55.3	43.0
	OPS	41.4	54.9	68.6	48.2	56.1	47.2	56.5	65.4	59.7	68.8	56.7
	SID	45.2	60.6	76.8	53.6	62.1	52.0	65.3	74.5	63.6	75.4	62.9
	EDA	53.6	71.7	80.9	64.5	73.2	63.1	74.4	80.0	72.4	82.1	71.6
IncRes-v2	L2T	42.3	54.6	71.9	48.8	55.5	48.6	59.6	63.1	58.1	67.2	57.0
	BSR	41.6	54.8	70.3	47.3	56.9	47.8	58.5	63.1	56.9	66.4	56.4
	DeCoWA	31.8	43.7	57.6	37.0	44.0	39.8	46.4	50.0	43.8	54.4	44.8
	OPS	43.2	56.1	69.1	48.9	57.6	49.5	60.5	61.3	55.3	65.7	56.7
	SID	50.7	67.6	78.7	55.0	65.9	58.0	68.6	73.7	64.1	72.3	65.5
	EDA	60.7	73.2	81.8	66.2	76.2	65.9	77.5	79.5	73.8	80.3	73.5

4.2.2. Using CNNs to attack ViTs

To evaluate the central cross-architecture setting, the same six attacks are applied to ten ViT-family targets. Table 2 presents the corresponding results, with EDA achieving the highest average ASR for every CNN source, exceeding the strongest source-specific baseline by 5.8, 11.2, 8.7, and 8.0 percentage points under RN-18, Inc-v3, Inc-v4, and IncRes-v2, respectively. The Inc-v4 source remains the most difficult of the four settings for EDA, with a ViT average of 71.6%; nevertheless, this is 8.7 percentage points above SID at 62.9%. The largest source-level gain occurs under Inc-v3, where EDA reaches 76.8% compared with 65.6% for the strongest baseline. At the target level, ViT-B and PiT-B are generally harder to attack than several hierarchical or hybrid Transformer variants. A plausible explanation is that sampling smooth deformations from both boundary-inclusive and interior control-point layouts reduces reliance on a single CNN-specific spatial configuration during gradient estimation. This interpretation is consistent with the intended deformation design; the component and layout ablations below examine the contributions of the corresponding factors rather than treating this architectural explanation as a causal proof.

4.2.3. Generalization beyond the ImageNet-compatible set

To determine whether the observed gains depend on the 1K ImageNet-Compatible benchmark, the evaluation is extended to a substantially larger in-distribution set and an independently collected distribution shift. The ImageNet-Compatible dataset (IN-C-1K) is retained as the reference benchmark.¹ ImageNet-V2 matched-frequency (IN-V2-10K) contains 10,000 independently collected images and represents a natural distribution shift.² The full ImageNet validation set (IN-Val-50K) provides a 50,000-image in-distribution evaluation.³

ASR is computed over the complete evaluation set as the percentage of adversarial examples whose top-1 prediction differs from the ground-truth label. Because clean accuracy may vary across datasets and target models, clean-correct ASR (CC-ASR) is also reported and computed

only over samples correctly classified before the attack. Thus, CC-ASR isolates attack-induced errors without changing the ASR definition used in the main experiments. Table 3 displays the cross-dataset results, with EDA achieving the highest ASR and CC-ASR for every evaluated source and target group on all three datasets. The full ImageNet validation set provides the most demanding scale test. On IN-Val-50K, EDA improves ASR over the strongest baseline by 0.4–2.7 percentage points on CNN targets and 3.2–7.1 points on ViT targets; the corresponding CC-ASR gains are 0.6–3.3 and 3.8–8.6 points. EDA also maintains the highest reported values on the distribution-shifted IN-V2-10K set. The persistence of the gains on both IN-Val-50K and IN-V2-10K indicates that the observed transfer improvement is not confined to the composition of the original 1K benchmark. The larger margins on ViT targets are also consistent with EDA's primary goal of reducing cross-architecture transfer degradation.

4.2.4. Transferability under different perturbation budgets and targeted settings

To test sensitivity to the perturbation budget and attack objective, four budgets are evaluated under both untargeted and targeted objectives. The tested values are $\epsilon \in \{\frac{4}{255}, \frac{8}{255}, \frac{12}{255}, \frac{16}{255}\}$. For untargeted attacks, the objective is to increase the classification loss with respect to the ground-truth label. For targeted attacks, the objective is reversed to encourage the adversarial example to be classified as the target class. Following the dataset protocol, the target class label provided for each image in the ImageNet-Compatible Dataset is kept identical across all methods, budgets, and source models. Targeted ASR is defined as the percentage of adversarial examples classified as the provided target class by the black-box target model. The number of iterations is fixed to $T = 10$, and the step size is scaled as $\alpha = \frac{\epsilon}{T}$. The untargeted results at $\epsilon = \frac{16}{255}$ reproduce the corresponding averages reported in the main comparison.

The results in Table 4 reveal two main trends: EDA achieves the highest untargeted transfer in every tested source-budget setting, while targeted transfer remains substantially more difficult for all methods. EDA nevertheless gives the best or tied-best targeted ASR in most settings, particularly at larger budgets. The most difficult regime is targeted transfer at small budgets, where ViT-target ASRs are near zero

¹ Available from the [Kaggle dataset page](#).

² Available from the [ImageNet-V2 repository](#).

³ See the [ILSVRC 2012 dataset page](#).

Table 3

Attack success rates (%) on the ImageNet-Compatible, ImageNet-V2 matched-frequency, and ImageNet validation datasets; each entry gives source-excluded averages for the CNN and ViT target groups.

Dataset	Attack	ASR				CC-ASR			
		RN-18	Inc-v3	Inc-v4	IncRes-v2	RN-18	Inc-v3	Inc-v4	IncRes-v2
IN-C-1K	L2T	95.2/72.0	<u>91.3/65.6</u>	85.4/55.5	85.4/57.0	95.0/70.9	<u>90.7/64.2</u>	84.5/53.8	84.4/55.3
	BSR	96.7/73.6	89.1/59.5	84.2/52.0	87.6/56.4	96.5/72.6	88.4/57.9	83.2/50.2	86.8/54.6
	DeCoWA	95.0/75.0	87.2/59.5	76.8/43.0	77.2/44.8	94.8/74.2	86.4/58.1	75.6/41.1	75.9/42.9
	OPS	95.0/75.4	87.9/59.7	85.1/56.7	83.5/56.7	94.8/74.4	87.2/58.2	84.1/55.1	82.3/55.1
	SID	<u>97.4/79.7</u>	90.3/62.4	<u>89.6/62.9</u>	<u>89.5/65.5</u>	<u>97.2/78.8</u>	89.6/60.9	<u>88.9/61.5</u>	<u>88.9/64.1</u>
	EDA	97.9/85.5	94.6/76.8	91.6/71.6	91.4/73.5	97.8/85.0	94.2/75.9	91.0/70.4	90.7/72.5
IN-V2-10K	L2T	96.9/85.9	95.0/84.0	91.7/78.8	91.6/79.4	95.2/80.4	<u>92.3/77.6</u>	87.0/70.3	86.8/71.0
	BSR	97.6/87.0	94.0/81.1	91.4/76.8	93.5/79.9	96.3/81.9	90.8/73.8	86.8/67.8	90.1/72.0
	DeCoWA	97.8/89.7	94.2/83.3	89.2/74.2	89.3/75.7	96.6/85.7	91.1/76.7	83.3/64.0	83.5/66.1
	OPS	97.8/89.5	94.7/82.8	92.5/80.2	91.6/80.4	96.7/85.4	91.9/76.1	88.4/72.4	86.9/72.5
	SID	<u>98.5/91.0</u>	<u>95.4/83.9</u>	<u>94.7/83.3</u>	<u>94.8/84.6</u>	<u>97.7/87.5</u>	<u>92.9/77.5</u>	<u>91.8/76.6</u>	<u>91.8/78.3</u>
	EDA	98.7/93.0	97.0/88.9	95.3/86.3	95.2/87.5	98.0/90.3	95.3/84.5	92.7/80.9	92.4/82.5
IN-Val-50K	L2T	94.6/78.8	91.8/76.0	87.2/69.8	86.6/70.4	93.1/74.5	89.3/71.1	83.3/63.7	82.5/64.2
	BSR	96.0/80.7	90.7/73.0	87.1/67.7	89.7/71.8	94.9/76.8	88.1/67.6	83.3/61.2	86.7/66.0
	DeCoWA	96.3/84.4	90.7/75.5	84.0/64.9	83.8/66.3	95.2/81.2	87.9/70.5	79.3/57.8	79.0/59.4
	OPS	96.1/83.6	91.3/74.6	88.2/71.9	86.8/71.7	95.0/80.2	88.8/69.4	84.7/66.1	82.8/65.9
	SID	<u>97.4/85.9</u>	<u>92.3/76.0</u>	<u>91.5/75.6</u>	<u>91.3/76.9</u>	<u>96.6/83.0</u>	<u>90.1/71.1</u>	<u>89.0/70.6</u>	<u>88.7/72.2</u>
	EDA	97.8/89.1	95.0/83.1	92.4/79.7	92.1/81.3	97.2/86.8	93.4/79.7	90.1/75.5	89.6/77.4

Table 4

Transfer ASR (%) under four perturbation budgets and two attack objectives; each entry gives source-excluded averages for the CNN and ViT target groups, and rankings are based on unrounded values.

Source	ϵ	Untargeted ASR (CNNs/ViTs)						Targeted ASR (CNNs/ViTs)					
		L2T	BSR	DeCoWA	OPS	SID	EDA	L2T	BSR	DeCoWA	OPS	SID	EDA
RN-18	4/255	45.8/19.8	53.8/22.8	49.1/22.7	49.4/21.3	<u>54.6/24.2</u>	57.3/27.0	<u>0.4/0.0</u>	0.3/0.0	0.3/0.0	0.3/0.0	<u>0.4/0.0</u>	0.5/0.0
	8/255	76.4/41.2	82.8/46.3	78.0/45.2	78.1/43.9	<u>83.9/49.3</u>	85.8/56.8	<u>4.0/0.4</u>	4.4/0.3	2.6/0.3	3.1/0.4	<u>4.5/0.4</u>	5.6/0.7
	12/255	89.6/59.9	93.3/63.6	89.8/63.6	90.2/62.9	<u>94.0/68.2</u>	95.1/75.8	9.4/1.5	10.3/1.4	6.5/1.2	7.5/1.3	<u>12.0/1.9</u>	14.3/3.1
	16/255	95.2/72.0	96.7/73.6	95.0/75.0	95.0/75.4	<u>97.4/79.7</u>	97.9/85.5	14.0/3.1	16.1/3.2	10.5/2.5	11.9/3.1	<u>17.4/4.2</u>	21.4/6.5
Inc-v3	4/255	35.5/14.9	<u>39.9/16.5</u>	37.0/15.6	36.1/14.9	39.0/15.9	45.6/20.2	0.0/0.0	<u>0.0/0.0</u>	0.0/0.0	0.0/0.0	0.0/0.0	0.1/0.0
	8/255	65.6/34.0	<u>68.8/34.3</u>	65.4/33.0	63.4/31.8	<u>68.4/34.8</u>	77.2/46.1	0.6/0.0	<u>0.8/0.0</u>	0.5/0.1	0.5/0.0	0.5/0.0	1.2/0.1
	12/255	83.4/52.4	82.7/49.2	80.2/48.8	79.8/48.0	<u>83.6/51.1</u>	89.7/65.9	2.1/0.3	<u>2.8/0.4</u>	1.9/0.3	1.6/0.2	2.2/0.4	3.7/0.8
	16/255	<u>91.3/65.6</u>	89.1/59.5	87.2/59.5	87.9/59.7	90.3/62.4	94.6/76.8	4.0/0.9	<u>5.4/1.1</u>	3.8/1.0	3.2/0.8	4.4/1.0	6.8/2.1
Inc-v4	4/255	34.4/14.9	40.0/17.2	32.2/13.6	36.6/15.8	<u>44.4/19.8</u>	45.9/22.2	0.0/0.0	<u>0.0/0.0</u>	0.0/0.0	0.0/0.0	0.1/0.0	0.0/0.0
	8/255	59.8/30.7	64.1/31.4	54.1/24.0	61.4/31.3	<u>69.9/38.4</u>	72.7/45.3	0.4/0.0	<u>0.3/0.0</u>	0.1/0.0	0.4/0.0	0.6/0.1	0.5/0.1
	12/255	76.5/45.4	77.2/43.5	68.3/34.6	77.0/45.4	<u>82.7/53.3</u>	86.0/61.5	1.3/0.2	<u>1.3/0.2</u>	0.5/0.1	1.1/0.2	1.7/0.2	1.5/0.3
	16/255	85.4/55.5	84.2/52.0	76.8/43.0	85.1/56.7	<u>89.6/62.9</u>	91.6/71.6	2.2/0.4	2.3/0.4	1.0/0.2	1.6/0.4	<u>2.9/0.5</u>	3.1/0.8
IncRes-v2	4/255	33.8/14.7	<u>42.4/18.6</u>	30.4/12.6	33.7/14.8	<u>41.9/19.1</u>	44.0/20.8	0.0/0.0	<u>0.1/0.0</u>	0.0/0.0	0.0/0.0	0.0/0.0	0.1/0.0
	8/255	58.6/30.2	67.4/34.6	52.0/24.0	58.7/29.4	<u>68.1/37.6</u>	71.5/45.1	0.5/0.0	<u>0.7/0.0</u>	0.2/0.0	0.3/0.0	<u>0.6/0.0</u>	0.6/0.1
	12/255	76.0/45.1	81.4/47.4	67.2/34.9	74.6/45.1	<u>82.4/54.1</u>	85.3/62.7	1.4/0.2	<u>1.7/0.2</u>	0.4/0.0	1.0/0.2	<u>1.9/0.3</u>	2.0/0.4
	16/255	85.4/57.0	87.6/56.4	77.2/44.8	83.5/56.7	<u>89.5/65.5</u>	91.4/73.5	2.5/0.5	2.7/0.4	1.0/0.1	1.9/0.5	<u>3.3/0.7</u>	3.4/1.0

for all attacks. Even at $\epsilon = \frac{16}{255}$, the targeted values remain far below their untargeted counterparts; for example, under RN-18, EDA reaches 21.4%/6.5% targeted ASR on CNN/ViT targets compared with 97.9%/85.5% in the untargeted setting. These results show that transformation diversity remains useful as the perturbation budget changes, but it does not remove the additional directional constraint imposed by targeted cross-model transfer. The experiment therefore supports robustness across budgets while preserving targeted transfer as a clear limitation.

4.2.5. Evaluation on defense models

To test whether the final perturbations remain effective after common defense or preprocessing operations, transfer is evaluated on adversarially trained, denoised, purified, and randomized-smoothed targets. Table 5 presents the defense-transfer results, with EDA obtaining the highest average ASR under all four CNN sources, although it is not uniformly best on every individual defense. The strongest resistance appears on several adversarially trained and randomized-smoothed targets. Under RN-18, for example, OPS slightly exceeds EDA on adv-RN-50(ℓ_∞) (38.7% versus 38.1%) and RS (48.1% versus 47.9%). Under Inc-

v4, OPS also exceeds EDA on RS (39.5% versus 36.9%). These cases illustrate that the average advantage should not be interpreted as uniform dominance over every defense.

EDA does not output a geometrically warped adversarial image; the deformation is used only while estimating gradients over transformed views. Optimizing one perturbation across these views can reduce dependence on an exact pixel-aligned pattern and is consistent with the observed resilience to several preprocessing operations. Randomized smoothing aggregates predictions of a base classifier over Gaussian-noised copies of the input (Cohen et al., 2019). EDA's gradient estimator samples spatially deformed views and also includes a Gaussian-noise appearance branch, so the perturbation is optimized under changes in pixel alignment and, for part of the transformed views, additive input noise. Although these transformations are not identical to the randomized-smoothing certification distribution, they provide a plausible explanation for the empirical RS transfer by reducing dependence on one exact input realization. This remains an empirical interpretation rather than evidence that EDA bypasses a certified radius. Recent perturbation-defense and feature-fusion studies further illustrate that robust evaluation remains an evolving setting (Wang et al., 2025; Xue et al., 2024a,b).

Table 5

Attack success rates (%) against the evaluated defended and preprocessing-based target models.

Source	Attack	Adversarial Training				Inception(timm)		Denoiser	Purifier Network		Smoothing	Avg.
		adv-RN-50(ℓ_2)	adv-RN-50(ℓ_∞)	FAT(2px)	FAT(4px)	Inc-v3 (timm)	IncRes-v2 (timm)		NRP(RN-101)	NRP(ViT-B)		
RN-18	L2T	53.4	35.7	50.7	50.9	69.6	48.6	87.3	49.8	23.8	41.9	51.1
	BSR	49.3	33.6	48.1	48.4	70.6	49.5	92.0	46.3	22.0	39.1	49.8
	De-CoWA	53.3	35.5	52.7	51.0	75.2	58.5	91.3	45.1	25.8	43.5	53.1
	OPS	60.9	38.7	56.2	54.0	79.9	65.7	90.2	53.7	28.5	48.1	57.5
	SID	59.2	37.4	55.2	<u>52.3</u>	<u>81.5</u>	<u>63.9</u>	<u>94.8</u>	<u>50.7</u>	<u>29.1</u>	47.0	57.1
	EDA	61.4	<u>38.1</u>	57.5	<u>52.3</u>	82.3	66.8	95.6	58.6	32.4	<u>47.9</u>	59.3
Inc-v3	L2T	45.5	34.6	47.4	48.8	73.8	57.1	89.2	<u>43.1</u>	23.9	39.9	50.3
	BSR	38.9	32.2	42.9	46.4	67.6	50.1	85.7	<u>34.6</u>	19.5	35.4	45.3
	De-CoWA	42.2	33.5	45.7	47.2	72.7	57.9	85.6	34.3	21.5	37.5	47.8
	OPS	<u>46.5</u>	35.5	<u>48.5</u>	49.6	76.7	<u>64.7</u>	87.8	41.8	23.1	<u>42.6</u>	51.6
	SID	46.0	34.4	46.5	48.5	<u>77.8</u>	64.6	<u>90.1</u>	40.4	<u>24.0</u>	40.4	51.2
	EDA	49.8	<u>34.8</u>	49.7	<u>49.2</u>	84.6	73.0	94.5	49.0	28.3	43.0	55.6
Inc-v4	L2T	39.6	32.8	43.8	46.3	68.5	53.8	79.6	33.4	18.5	35.8	45.2
	BSR	35.2	30.3	40.2	44.2	60.4	43.5	77.7	27.9	16.0	32.1	40.7
	De-CoWA	35.4	30.9	41.0	44.8	65.9	49.2	71.6	26.3	15.7	32.4	41.3
	OPS	<u>43.9</u>	33.9	<u>44.6</u>	47.4	74.4	64.5	82.2	<u>37.4</u>	20.6	39.5	48.8
	SID	44.2	32.6	45.3	<u>46.5</u>	<u>75.6</u>	<u>65.1</u>	<u>89.1</u>	35.2	<u>21.1</u>	36.8	<u>49.1</u>
	EDA	43.2	32.2	44.4	45.5	<u>77.2</u>	69.6	91.0	41.2	23.8	<u>36.9</u>	50.5
IncRes-v2	L2T	44.5	34.5	45.1	47.4	74.0	64.0	82.1	36.6	22.1	38.3	48.8
	BSR	38.7	31.6	42.0	45.6	70.6	58.1	85.3	32.0	18.6	34.2	45.6
	De-CoWA	40.3	31.7	43.1	45.9	73.7	61.5	76.5	28.8	19.0	36.2	45.6
	OPS	47.3	35.5	<u>48.6</u>	48.9	80.4	75.2	83.4	<u>39.1</u>	22.4	43.4	52.4
	SID	48.4	35.3	<u>47.6</u>	47.8	84.2	78.7	91.5	38.5	<u>24.1</u>	41.8	53.7
	EDA	50.3	<u>35.3</u>	49.0	<u>47.9</u>	<u>84.0</u>	80.2	92.8	46.5	29.0	41.6	55.7

4.3. Extended evaluations

Unless otherwise stated, all reported target-group averages follow the source-excluded averaging protocol described in the experimental settings.

4.3.1. Evaluation on modern robust models

To assess transferability against stronger and more recent robustness regimes, EDA is evaluated on eight ImageNet ℓ_∞ robust classifiers from current robustness benchmarks and recent robust-model families. The evaluation includes checkpoints from the RobustBench Model Zoo (Croce et al., 2021)⁴ and recent MIMIR- and MeanSparse-based models (Amini et al., 2024; Chen & Lee, 2024; Liu et al., 2025; Peng et al., 2023; Singh et al., 2023; Xu et al., 2026). The selected targets cover ConvNeXt-, ViT-, Swin-, Wide-ResNet-, RobustArch-, MIMIR-, and MeanSparse-based robust architectures. The exact ImageNet/ ℓ_∞ model identifiers are Singh2023Revisiting_ConvNeXt-T-ConvStem, Singh2023Revisiting_ViT-B-ConvStem, Liu2023Comprehensive_Swin-B, Chen2024Data_WRN_50_2, Peng2023Robust, Xu2024MIMIR_Swin-B, Liu2023Comprehensive_ConvNeXt-L, and Amini2024MeanSparse_ConvNeXt-L. Each target is evaluated with the checkpoint and preprocessing registered for the corresponding benchmark entry. Avg. is computed over the listed robust targets.

The transfer ASRs displayed in Table 6 are substantially lower than those on normally trained models. EDA achieves the highest displayed average under Inc-v3, Inc-v4, and IncRes-v2 and remains comparable to OPS under RN-18, for which both averages round to 20.4%. The robust-target setting exposes the clearest remaining transfer gap. Under Inc-v4, for example, EDA reaches 14.1% on RB-ViT-B, below OPS at 15.3%, even though EDA has the highest overall average for that source. The

low absolute ASRs across all methods demonstrate that current robust classifiers are substantially harder to transfer to than normally trained targets. The mixed target-wise ranking suggests that stochastic deformation can preserve some transferable directions but cannot eliminate the gradient and decision-boundary changes induced by robust training. This experiment therefore establishes the robustness-oriented scope of the evaluation rather than demonstrating that EDA is best on the robust model.

4.3.2. Evaluation with ViT source models

To determine whether EDA depends on using a CNN surrogate, ViT-B and LeViT are additionally used as source models. Table 7 illustrates EDA's source-dependent behavior: it ranks second to SID for both target groups under ViT-B and ranks first for both groups under LeViT. Following the averaging protocol, a source-identical target is excluded whenever the source model appears in the corresponding target group. The ViT-B source is the harder case for demonstrating an advantage: EDA reaches 86.8% on CNN targets and 92.4% on ViT targets, compared with SID at 87.4% and 93.5%. With LeViT as the source, EDA reaches 96.8% and 96.9%, marginally exceeding SID in both groups. This source dependence is consistent with EDA's intended setting. A ViT surrogate already shares patch-based or global aggregation behavior with many ViT targets, so the CNN-to-ViT mismatch that motivates EDA is less pronounced and the baseline transfer level leaves less room for improvement. The experiment therefore supports applicability beyond CNN surrogates without claiming uniform gains in that regime.

4.3.3. Stability across random seeds

Because EDA samples the control-point layout, TPS offsets, and appearance transformation, stability across stochastic realizations is evaluated. For each base seed in {0, 1, 2, 3, 4}, adversarial examples are regenerated from scratch while the dataset, source and target models, perturbation budget, iteration number, and attack hyperparameters are kept fixed. The main single-run tables use seed 42 and are not included in this five-seed summary; all runs in this stability analysis use $\epsilon = \frac{16}{255}$. Table 8

⁴ Official RobustBench resources are available at <https://github.com/RobustBench/robustbench>.

Table 6Attack success rates (%) against eight modern ImageNet ℓ_∞ robust classifiers; Avg. is computed over the listed targets.

Source	Attack	Modern ImageNet ℓ_∞ Robust Models								Avg.
		RB-ConvNeXt-T	RB-ViT-B	RB-Swin-B	RB-WRN-50-2	RB-RobustArch	MIMIR-Swin-B	RB-ConvNeXt-L	MeanSparse-CNX-L	
RN-18	L2T	20.3	16.4	15.4	30.6	22.8	15.7	14.3	14.6	18.8
	BSR	18.4	15.3	14.6	27.5	20.4	14.0	12.9	12.9	17.0
	DeCoWA	20.2	16.8	15.3	30.0	22.1	15.5	14.5	14.9	18.7
	OPS	22.8	18.4	<u>17.0</u>	<u>32.5</u>	<u>23.9</u>	<u>17.4</u>	<u>15.4</u>	15.9	20.4
	SID	21.8	17.9	16.7	31.5	23.2	16.1	15.1	15.2	19.7
	EDA	<u>22.2</u>	18.4	17.1	32.8	24.2	<u>16.7</u>	15.7	<u>15.8</u>	20.4
Inc-v3	L2T	21.5	15.3	15.3	28.5	21.7	15.6	13.6	14.1	18.2
	BSR	19.3	13.3	13.7	26.1	19.5	14.2	12.5	13.1	16.5
	DeCoWA	19.6	14.2	14.1	26.6	19.7	14.3	13.0	13.4	16.9
	OPS	<u>21.9</u>	<u>16.9</u>	15.8	<u>29.3</u>	<u>22.0</u>	<u>17.5</u>	<u>14.3</u>	<u>14.7</u>	<u>19.1</u>
	SID	20.9	16.0	<u>16.1</u>	29.2	<u>22.0</u>	15.9	13.9	14.5	18.6
	EDA	22.5	17.2	17.4	30.0	22.9	17.8	15.4	15.5	19.8
Inc-v4	L2T	18.5	14.6	13.4	26.0	20.4	14.4	12.9	13.5	16.7
	BSR	16.5	12.4	12.2	24.3	17.8	13.0	11.1	11.6	14.9
	DeCoWA	17.4	12.5	12.7	24.7	18.4	13.0	12.4	12.1	15.4
	OPS	20.7	15.3	<u>14.8</u>	<u>27.6</u>	20.2	<u>14.9</u>	<u>13.5</u>	<u>14.1</u>	<u>17.6</u>
	SID	19.0	<u>15.1</u>	14.2	27.1	20.9	14.1	13.0	13.3	17.1
	EDA	<u>20.4</u>	14.1	15.2	27.8	20.0	15.8	14.9	15.0	17.9
IncRes-v2	L2T	20.3	15.9	15.5	27.8	21.4	16.2	15.0	15.2	18.4
	BSR	18.2	13.7	13.6	26.8	19.4	14.1	12.7	12.7	16.4
	DeCoWA	19.4	14.1	14.1	27.3	20.8	13.7	13.6	13.6	17.1
	OPS	<u>22.1</u>	17.5	16.6	29.3	<u>22.6</u>	<u>17.2</u>	15.0	15.4	<u>19.5</u>
	SID	21.7	<u>17.4</u>	16.3	<u>30.1</u>	21.9	15.9	<u>15.9</u>	<u>15.9</u>	19.4
	EDA	22.8	<u>17.4</u>	16.9	30.2	23.1	17.4	16.8	<u>17.4</u>	20.2

Table 7

Source-excluded transfer ASR (%) from ViT-B and LeViT surrogates to CNN and ViT target groups.

Source	Target group	Attack					
		L2T	BSR	DeCoWA	OPS	SID	EDA
ViT-B	CNNs	85.3	86.2	82.7	86.4	87.4	<u>86.8</u>
	ViTs	90.1	91.6	87.0	89.6	93.5	<u>92.4</u>
LeViT	CNNs	93.2	95.5	93.1	93.7	<u>96.7</u>	96.8
	ViTs	91.4	92.7	90.7	92.0	<u>96.8</u>	96.9

demonstrates that EDA has the highest mean ASR in all eight combinations of source model and target group and small sample standard deviations across the five evaluated seeds. The largest EDA standard deviation is 0.60 percentage points for the IncRes-v2-to-ViT setting, where the mean remains 73.01%, well above SID at 65.02%. Thus, the ranking advantage is not explained by one favorable seed in the evaluated set. This stability is consistent with averaging K independently sampled transformation-path gradients at each iteration, which reduces the influence of any single transformed view. The five-seed study supports stability over the tested seeds but is not used to claim seed-independence beyond them.

4.3.4. Perceptual quality of final adversarial examples

To verify that optimization-time geometric warping does not introduce substantial additional distortion, perceptual quality is evaluated between the clean image x and the final adversarial output x^{adv} under the RN-18 source. This output remains in the original coordinate system and satisfies the same ℓ_∞ constraint as the compared baselines. SSIM, PSNR, LPIPS, NMSE, NLPD, and GMSD are computed between x and x^{adv} , while TV is computed on $\delta = x^{\text{adv}} - x$. Table 9 presents comparable values for EDA and the baselines across the seven metrics, rather than indicating that EDA is best on every individual measure. The most informative comparison is therefore the absence of a systematic quality penalty. EDA obtains LPIPS 0.3058 and TV 0.0638, close to the best values of 0.2969 and 0.0616, while its PSNR of 26.57 remains near the

26.57–26.77 range of the baselines. Its NMSE, NLPD, and GMSD are likewise within the range of the compared attacks. This behavior follows from the attack construction: TPS warping is applied only to intermediate views used for gradient estimation, whereas the final output is an additive, ℓ_∞ -bounded adversarial example in the original geometry. The perceptual experiment therefore checks the consequence of the optimization procedure without implying that EDA optimizes these quality metrics directly.

4.3.5. Combination with existing gradient-based attacks

To assess implementation compatibility rather than claim a separate source of novelty, EDA is integrated into VMI-FGSM (Wang & He, 2021), EMI-FGSM (Wang et al., 2021b), PGN (Ge et al., 2023), MEF (Qiu et al., 2026), and GAA (Gan & Wo, 2025). For each baseline, its original gradient optimization rule is kept unchanged, and EDA is used only as the input-transformation module during gradient estimation. Table 10 demonstrates that adding EDA increases every reported CNN and ViT group average under the RN-18 source. The improvement is not limited to weak baselines. MEF already has the strongest baseline Avg. of 74.9%, yet adding EDA raises it to 81.0% (+ 6.1 points). The largest average gain occurs for PGN, from 68.6% to 84.8% (+ 16.2 points), while GAA gains 22.4 points on ViT targets. Because the original optimization rules are retained and only the transformed inputs used for gradient estimation are changed, these gains establish implementation compatibility with existing gradient-based frameworks. They do not constitute an additional novelty claim beyond the deformation design itself.

4.3.6. Feature-space and attention-space analysis

To examine whether the transfer gains are accompanied by changes in the target model's internal representation and spatial evidence, Swin-T (Liu et al., 2021) is analyzed using final-encoder features and Grad-CAM maps. Let $h(\cdot)$ denote the final encoder output, $p = \text{softmax}(\text{vec}(h(x)))$, and $q = \text{softmax}(\text{vec}(h(x^{\text{adv}})))$. The reported KL metric is $D_{\text{KL}}(q||p) = \sum_j q_j \log \frac{q_j}{p_j}$, computed with a small numerical stabilizer. In feature space, L_2 distance, KL divergence, and cosine similarity respectively measure shift magnitude, redistribution of softmax-normalized flattened responses, and directional alignment. For

Table 8Attack success rates (%) over seeds {0, 1, 2, 3, 4}, reported as mean $\pm$ sample standard deviation.

Attack	RN-18		Inc-v3		Inc-v4		IncRes-v2	
	CNNs	ViTs	CNNs	ViTs	CNNs	ViTs	CNNs	ViTs
L2T	95.16 $\pm$ 0.12	72.04 $\pm$ 0.28	90.48 $\pm$ 0.18	66.24 $\pm$ 0.31	85.38 $\pm$ 0.22	55.46 $\pm$ 0.34	85.42 $\pm$ 0.25	56.82 $\pm$ 0.29
BSR	96.74 $\pm$ 0.10	73.70 $\pm$ 0.25	89.02 $\pm$ 0.21	60.08 $\pm$ 0.08	84.29 $\pm$ 0.19	51.67 $\pm$ 0.39	87.75 $\pm$ 0.23	56.89 $\pm$ 0.48
DeCoWA	95.07 $\pm$ 0.10	75.21 $\pm$ 0.32	87.53 $\pm$ 0.24	60.23 $\pm$ 0.32	76.33 $\pm$ 0.33	43.20 $\pm$ 0.69	76.56 $\pm$ 0.43	44.42 $\pm$ 0.20
OPS	94.93 $\pm$ 0.13	75.07 $\pm$ 0.43	88.14 $\pm$ 0.39	59.89 $\pm$ 0.48	84.54 $\pm$ 0.40	56.05 $\pm$ 0.21	83.41 $\pm$ 0.40	56.32 $\pm$ 0.30
SID	97.27 $\pm$ 0.08	79.65 $\pm$ 0.11	90.26 $\pm$ 0.19	62.02 $\pm$ 0.21	89.16 $\pm$ 0.18	62.67 $\pm$ 0.17	89.26 $\pm$ 0.18	65.02 $\pm$ 0.33
EDA	97.92 $\pm$ 0.10	84.96 $\pm$ 0.18	94.44 $\pm$ 0.05	76.52 $\pm$ 0.43	91.54 $\pm$ 0.25	70.86 $\pm$ 0.38	91.62 $\pm$ 0.39	73.01 $\pm$ 0.60

Table 9

Perceptual quality of the final adversarial examples generated with the RN-18 source.

Attack	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	TV $\downarrow$	NMSE $\downarrow$	NLPD $\downarrow$	GMSD $\downarrow$
L2T	0.6878	26.71	0.3171	0.0750	0.0104	0.2704	0.1667
BSR	0.7024	26.77	0.3090	0.0646	0.0102	0.2686	0.1700
DeCoWA	0.7007	26.70	0.2969	0.0647	0.0104	0.2731	0.1686
OPS	0.6945	26.57	0.3181	0.0706	0.0107	0.2904	0.1642
SID	0.7043	26.66	0.3123	0.0616	0.0105	0.2787	0.1699
EDA	0.6985	26.57	0.3058	0.0638	0.0107	0.2834	0.1676

Table 10

Compatibility of EDA with five gradient-based attacks under the RN-18 source; superscripts give absolute ASR gains in percentage points after adding EDA.

Attack	CNNs	ViTs	Avg.
VMI-FGSM	81.2	50.0	65.6
+ EDA	89.6 ^{+8.4}	65.3 ^{+15.4}	77.5 ^{+11.9}
EMI-FGSM	81.9	44.6	63.3
+ EDA	88.3 ^{+6.4}	58.6 ^{+14.0}	73.5 ^{+10.2}
PGN	85.0	52.2	68.6
+ EDA	94.1 ^{+9.1}	75.4 ^{+23.2}	84.8 ^{+16.2}
MEF	89.7	60.2	74.9
+ EDA	92.1 ^{+2.4}	70.0 ^{+9.8}	81.0 ^{+6.1}
GAA	82.6	47.6	65.1
+ EDA	92.4 ^{+9.8}	70.0 ^{+22.4}	81.2 ^{+16.1}

Table 11

Feature-space and Grad-CAM attention differences between clean and adversarial inputs on Swin-T under the RN-18 source.

Attack	Feature map			Grad-CAM		
	$L_2(\uparrow)$	KL($\uparrow$)	Cosine($\downarrow$)	$L_2(\uparrow)$	JS($\uparrow$)	Corr($\downarrow$)
L2T	272.026	1.882	0.284	61.044	0.230	0.312
BSR	274.325	1.913	0.273	61.138	0.236	0.294
DeCoWA	271.918	1.887	0.279	61.039	0.235	0.296
OPS	269.929	1.833	0.281	60.517	0.230	0.319
SID	274.016	1.908	0.267	61.256	0.240	0.282
EDA	278.012	1.931	0.242	61.534	0.244	0.278

Grad-CAM, L_2 distance and JS divergence quantify attention shift, whereas correlation measures spatial similarity. Table 11 displays EDA with the largest feature-space L_2 distance and KL divergence and the smallest cosine similarity; it also has the largest Grad-CAM L_2 and JS divergence and the lowest correlation among the compared attacks. The margins on some metrics are modest—for example, KL is 1.931 for EDA versus 1.913 for BSR, and Grad-CAM correlation is 0.278 versus 0.282 for SID—so these measurements should be interpreted as complementary evidence rather than a standalone explanation of ASR. A larger KL value indicates stronger redistribution of the normalized final-encoder responses, while the Grad-CAM differences indicate a change in spatial evidence. These observations are consistent with EDA perturbing both feature and attention patterns through optimization over diverse trans-

Table 12

Component ablation under the RN-18 source; entries give source-excluded CNN ASR, ViT ASR, and their equally weighted average.

Adaptive expansion	Dual-layout switching	Appearance augmentation	Target model		Avg.
			CNNs	ViTs	
✓			97.2	78.0	87.6
			97.4	81.8	89.6
			97.4	78.0	87.7
✓	✓		97.8	81.2	89.5
			97.5	82.0	89.8
✓	✓	✓	97.8	84.8	91.3
			97.7	81.8	89.8
✓	✓	✓	97.9	85.5	91.7

formed views. However, the analysis is limited to one RN-18-to-Swin-T setting and does not prove that these representation changes causally produce the transfer gains or guarantee decision-boundary crossing.

4.4. Ablation study

The following experiments isolate the roles of EDA's spatial and appearance components and examine the sensitivity of their key settings.

4.4.1. Ablation of EDA components

To isolate the individual and combined contributions of adaptive expansion, dual-layout switching, and appearance augmentation, a component ablation is conducted under the RN-18 source. CNN averages follow the source-excluded protocol, and Avg. is the equally weighted mean of the reported CNN and ViT averages. Table 12 presents the component-wise results, with the full configuration achieving the highest overall average of 91.7%. The component effects are unequal and non-additive, with the clearest differences appearing on ViT targets. Cross-architecture transfer to ViTs provides the most discriminative comparison. The base configuration reaches 78.0% ViT ASR; expansion alone raises this to 81.8% and appearance augmentation alone to 81.2%, whereas dual-layout switching alone remains at 78.0%. Combining expansion with layout switching reaches 82.0%, and combining expansion with appearance augmentation reaches 84.8%. The full method reaches 85.5% while maintaining 97.9% source-excluded CNN ASR. The spatial-only configuration is also 7.0 percentage points above vanilla DeCoWA at 75.0% in Table 2, illustrating that the cross-architecture gain is not attributable solely to appearance variation. These controlled comparisons do not support treating the three components as equally strong independent contributors. Instead, the results suggest complementary but interacting roles: adaptive expansion provides boundary support, dual-layout switching varies the control-point geometry, and appearance augmentation introduces additional photometric diversity. Their combination yields the strongest overall performance, indicating that EDA benefits from jointly diversifying spatial and appearance transformations rather than from any single component alone.

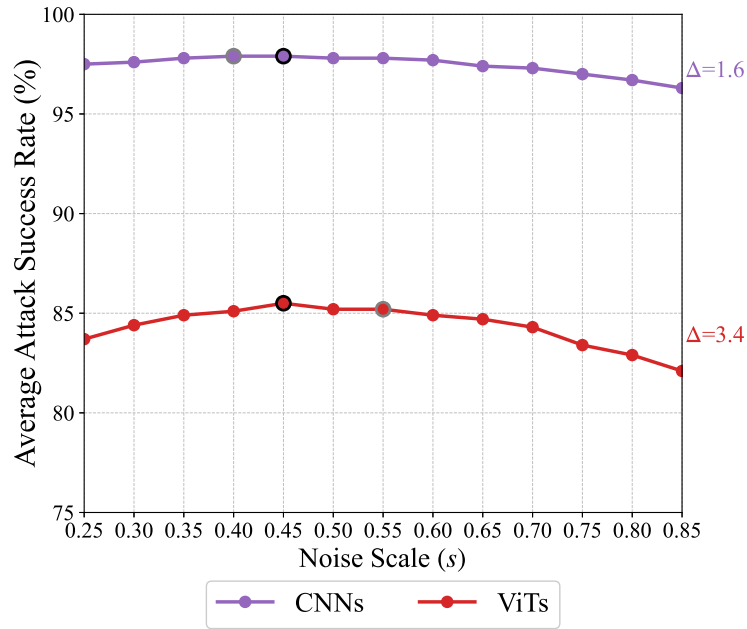


Fig. 7. Sensitivity of source-excluded CNN and ViT ASR to the deformation scale s under the RN-18 source.

Table 13

Control-point layout and full-grid boundary-movement ablation; each entry gives source-excluded ASRs for the CNN and ViT target groups under the indicated source model.

Layout modes	Source model			
	RN-18	Inc-v3	Inc-v4	IncRes-v2
full grid, fixed boundary	96.7/80.6	89.0/66.5	86.5/59.6	88.8/64.1
full grid, movable boundary	<u>97.8/84.8</u>	<u>93.6/74.9</u>	<u>91.0/69.5</u>	<u>91.4/71.3</u>
dual layout, movable boundary	<u>97.9/85.5</u>	<u>94.6/76.8</u>	<u>91.6/71.6</u>	<u>91.4/73.5</u>
dual layout, fixed boundary	<u>97.8/85.1</u>	<u>93.3/75.3</u>	<u>90.6/68.6</u>	<u>91.0/72.6</u>

4.4.2. Sensitivity to the noise scale

To determine how much geometric deformation is useful before spatial variation becomes excessive, the deformation scale s , which controls the sampled control-point offsets, is varied. Fig. 7 illustrates that the best tested setting is $s = 0.45$, while a stable high-performance region occurs around $s \in [0.40, 0.55]$ with less than 0.2 percentage points of variation in the transfer averages. ViT targets are more sensitive to this parameter than CNN targets across the tested range. When s becomes too large, ASR decreases for both model families, whereas nearby settings around 0.45 remain similar. The small difference between $s = 0.45$ and $s = 0.55$ should therefore not be over-interpreted. This trend is consistent with a balance between deformation diversity and preservation of useful gradient alignment: moderate displacements expose the surrogate to varied spatial views, while excessive displacement can weaken the correspondence needed for transferable gradients. The sweep supports $s = 0.45$ as the best tested setting, not as a theoretically optimal value.

4.4.3. Ablation of control point layout modes

To separate the effect of changing control-point geometry from the effect of moving full-grid boundary points, full-grid-only and dual-layout configurations with fixed or movable boundaries are compared. Table 13 presents a consistent pattern: the dual layout with movable full-grid boundaries gives the highest ViT average for all four CNN sources, and boundary movement improves ViT transfer in both matched full-grid-only and dual-layout comparisons. The Inc-v4 source gives the clearest boundary-movement contrast. In the full-grid-only setting, making the boundary movable raises ViT ASR from 59.6% to 69.5% (+9.9 points); within the dual layout, it raises the value from 68.6% to

71.6% (+3.0 points). Holding boundary movement fixed and adding the center-grid branch also improves the ViT average: for the movable-boundary rows, the gain from full grid to dual layout ranges from 0.7 to 2.2 points across the four sources. These matched comparisons separate two design effects. Boundary movement enlarges the deformation support of the full-grid branch, whereas the center-grid branch introduces an alternative interior control-point geometry. Their combination yields the strongest cross-architecture result without implying that boundary movement is a separate transformation module.

4.4.4. Ablation of expansion modes

To determine how the content outside the original image support affects TPS-based transfer, zero, Gaussian-noise, border, and reflection expansion are compared under the RN-18 source. Fig. 8 presents the comparison, in which the four modes are nearly indistinguishable on CNN targets, whereas reflection padding gives the highest ViT ASR of 85.5% and the highest equally weighted average of 91.7%. The cross-architecture result is the discriminative case: zero and Gaussian-noise padding obtain 82.0% and 81.4% ViT ASR, border padding reaches 84.9%, and reflection reaches 85.5%. By contrast, the CNN values remain within 97.7–97.9%. This pattern supports the use of reflection expansion as a boundary-handling choice. Reflection preserves local image continuity better than zero or random-noise filling and therefore reduces abrupt artificial boundary patterns in the intermediate warped views. The experiment establishes an empirical advantage for the tested padding modes rather than a general proof that reflection is optimal for all deformations.

4.4.5. Ablation and sensitivity analysis of appearance augmentations

To determine whether appearance augmentation contributes beyond the spatial deformation and to justify its parameter choices, component-removal tests are combined with sweeps over Gaussian-noise strength and brightness range. Fig. 9 demonstrates that both appearance operators contribute in the RN-18 setting, with brightness variation producing the larger removal penalty, and that the selected $\sigma = 0.05$ and $\mathcal{U}(0, 2)$ brightness range give the best tested combination. Removing Gaussian noise reduces ViT ASR from 85.5% to 83.5%, whereas removing brightness adjustment reduces it further to 82.8%. The sensitivity sweep provides the harder parameter test: large Gaussian-noise levels reduce transferability, especially on ViT targets, while widening the bright-

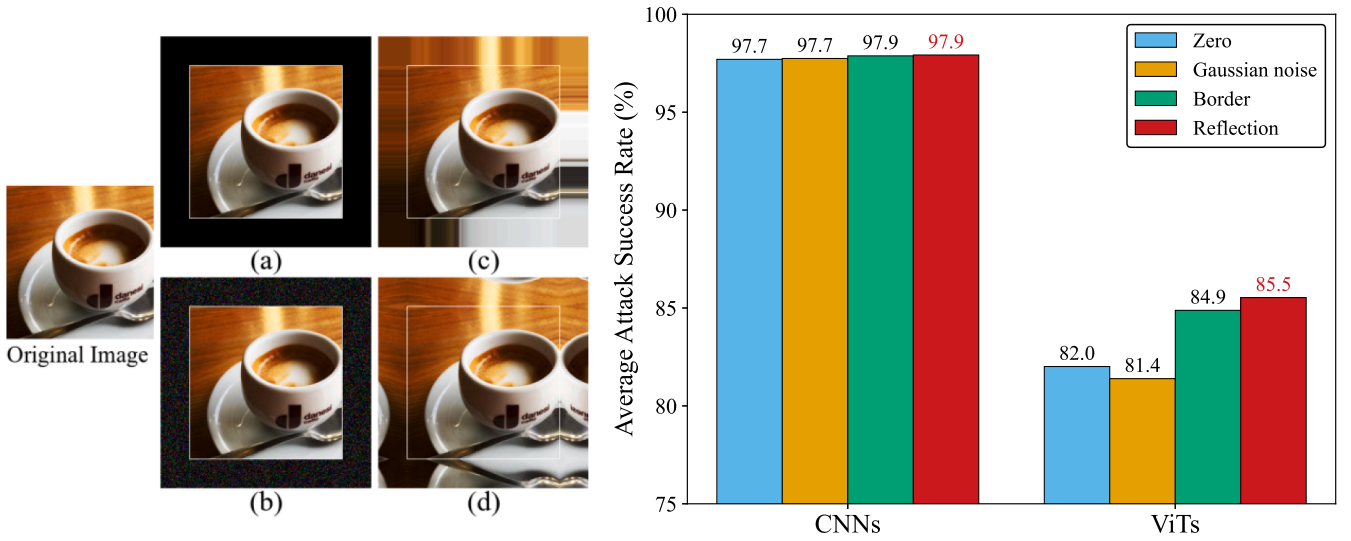


Fig. 8. Boundary-expansion ablation under the RN-18 source: image panels depict (a) zero, (b) Gaussian-noise, (c) border, and (d) reflection padding, and the bar chart gives source-excluded CNN and ViT ASR.

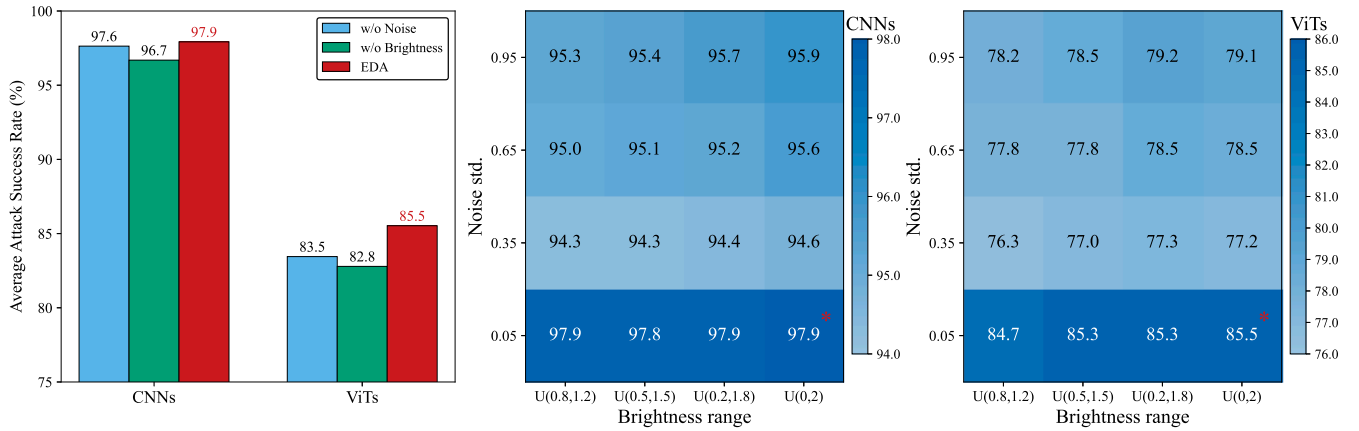


Fig. 9. Appearance-augmentation analysis under the RN-18 source: component ablation (left), sensitivity to the Gaussian-noise standard deviation (middle), and sensitivity to the brightness range (right).

ness interval with $\sigma = 0.05$ fixed improves the tested ViT result. These trends are consistent with complementary photometric roles. Weak pixel-level noise adds local appearance variation without overwhelming the surrogate gradient, while brightness adjustment varies image-level intensity. The selected settings are therefore justified by the reported sweep and are not claimed to be theoretically optimal or universally transferable.

4.5. Adversarial examples and perturbation maps

To verify visually that EDA's optimization-time deformations do not turn the final output into a warped image, representative adversarial examples and their perturbation maps are inspected. Fig. 10 visualizes two cases in which EDA changes the ViT-B prediction while the final adversarial examples retain the original image geometry and satisfy the same ℓ_∞ perturbation protocol. Each perturbation map is obtained by subtracting the clean image from the corresponding final adversarial example, with pixel brightness indicating perturbation intensity. The displayed probabilities are ViT-B probabilities for the indicated classes. Because geometric warping is used only inside gradient estimation, the visible output remains an additive perturbation in the original coordinate system. These examples provide qualitative support for the

implementation claim that deformation is an optimization-time operation rather than a direct geometric modification of the delivered adversarial image. They are illustrative cases and are not used as a substitute for the quantitative transfer and perceptual-quality evaluations.

4.6. Quantitative evaluation on VLMs

To examine whether transferred adversarial examples also degrade label-conditioned recognition beyond closed-set image classifiers, a quantitative evaluation is conducted on vision-language models (VLMs). The protocol uses paired clean and adversarial inputs so that each model is compared on the same valid image set and prompt conditions. A fixed subset of 100 ImageNet images is selected deterministically as the first 100 filenames after ascending filename sorting. For each image, the clean input and six adversarial counterparts generated by L2T, BSR, DeCoWA, OPS, SID, and EDA are evaluated. All adversarial examples use RN-18 as the source model. The evaluation covers 12 VLMs from seven model families. Within each VLM, the same model-specific valid image set, ground-truth labels, and prompts are used for the clean input and all attack methods. If an ImageNet class contains multiple aliases, the complete comma-separated label string is retained. The fixed prompt is as follows:

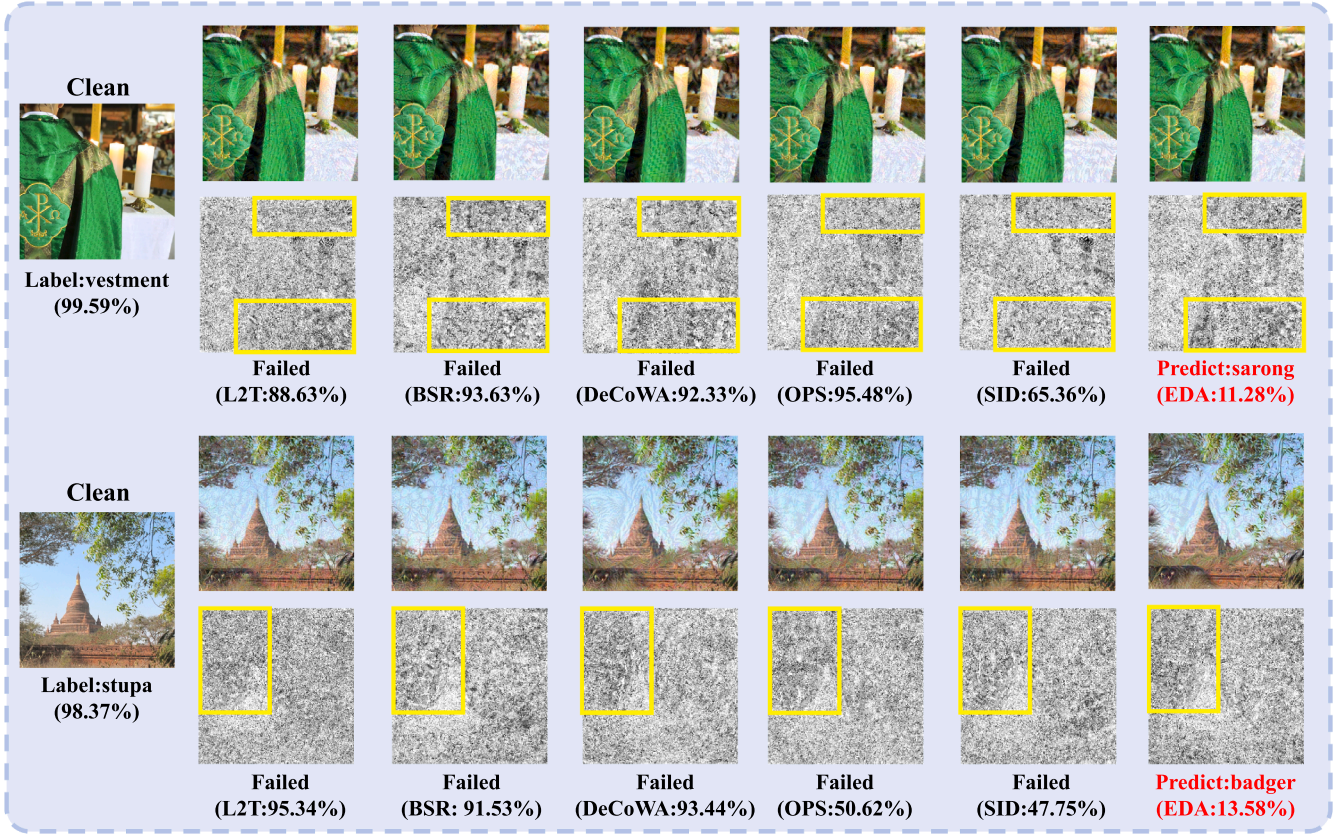


Fig. 10. Adversarial examples and perturbation maps produced by five baselines and EDA from two clean inputs; labels give ViT-B predictions and probabilities, and yellow boxes identify the displayed perturbation regions.

You are a strict image classification expert system.

Task:

Determine whether the main visual content of the given image belongs to the following label:

Label: <ground-truth ImageNet label>

Instructions:

1. Judge only from the visual content of the image.
2. Return "match": 1 if the main object or scene clearly matches the given label.
3. Return "match": 0 if the image does not match the given label.
4. Return a confidence score from 1 to 10.
5. Do not output any text other than the JSON object.

Required JSON format:

```
{"match": 1 or 0, "confidence": integer between 1 and 10}
```

For the API-accessed models, the provider-exposed identifiers are Claude (*claude-opus-4-8* and *claude-sonnet-5*), Gemini (*gemini-3.1-pro-preview* and *gemini-3.5-flash*), Qwen (*qwen3.7-plus* and *qwen3.6-plus*), and Kimi (*kimi-k2.7-code* and *kimi-k2.6*). GPT-5.6 Sol and GPT-5.5 High are accessed through isolated Codex tasks, whereas Grok 4.5 High and GLM 5.2 Max are accessed through isolated Cursor tasks. Five evaluations are collected for each valid model-input-condition-image combination. For API-accessed models, the five evaluations use the prespecified temperatures of 0.65, 0.80, 0.95, 1.10, and 1.25, with top- $p = .9$ when supported. For client-mediated interfaces, five isolated evaluations are conducted under provider-default settings for parameters

not exposed by the interface. The resulting standard deviations therefore summarize variability across these access-specific evaluation conditions rather than homogeneous same-temperature repeatability across all models. For input condition a , image i , VLM v , and evaluation t , the label-conditioned score is

$$s_{a,i,v}^{(t)} = m_{a,i,v}^{(t)} \frac{c_{a,i,v}^{(t)}}{10},$$

where $m_{a,i,v}^{(t)} \in \{0, 1\}$ is the match decision and $c_{a,i,v}^{(t)} \in \{1, \dots, 10\}$ is the reported confidence. Let I_v denote the model-specific valid-image set used for every input condition and evaluation of VLM v , and let $N_v = |I_v|$. The VLM-level percentage score is

$$S_{a,v}^{(t)} = \frac{100}{N_v} \sum_{i \in I_v} s_{a,i,v}^{(t)}.$$

The factor 100 converts the valid-image mean to a percentage; it is not an assumed sample count. Thus, the marked 99-image columns use the factor 100/99, whereas the remaining columns use 100/100. For each pair (a, v) , the mean and sample standard deviation of $S_{a,v}^{(t)}$ over the five evaluations are reported. The final Avg. column reports the unweighted arithmetic mean of the 12 VLM-wise mean scores. A lower adversarial score indicates weaker recognition of the original ground-truth category.

Under this paired protocol, Table 14 displays the paired scores, with EDA producing the lowest mean recognition score for 11 of the 12 VLMs and the lowest macro-average, 50.6%, compared with 53.7% for SID and 82.1% for clean images. The only model on which EDA is not the lowest-scoring attack is GPT-5.6 Sol, where SID is lower by 0.4 percentage points. Relative to the clean macro-average, EDA reduces the score by 31.5 points, while its standard deviations range from 0.4 to 2.6 points across the prescribed access-specific conditions. Because clean

Table 14
Label-conditioned recognition scores (%) across 12 VLMs, reported as mean $\pm$ sample standard deviation over five evaluations; lower adversarial scores indicate weaker recognition of the ground-truth category.

Method	GPT	Claude	Gemini	Qwen	Kimi	Grok	GLM	Avg. ↓					
	5.6 Sol	5.5 High	Sonnet 5	3.1 Pro-Preview	3.5 Flash	3.7 Plus ^a	3.6 Plus ^a	K2.6 ^a	K2.7-Code ^a	K2.6 ^a	4.5 High	5.2 Max	
Clean	90.1 ± 1.4	90.1 ± 0.6	79.9 ± 0.1	78.7 ± 0.3	82.5 ± 1.0	85.7 ± 0.5	81.6 ± 0.7	85.7 ± 0.4	80.8 ± 2.1	81.8 ± 0.5	73.4 ± 1.2	74.6 ± 2.3	82.1
L2T	78.5 ± 1.3	77.6 ± 1.3	36.0 ± 0.4	40.6 ± 0.5	58.9 ± 2.1	71.4 ± 0.6	49.9 ± 1.3	62.9 ± 1.1	55.6 ± 1.6	50.7 ± 2.1	56.8 ± 2.6	47.2 ± 1.5	57.2
BSR	77.8 ± 1.2	76.2 ± 1.1	38.2 ± 0.3	43.6 ± 0.4	60.9 ± 1.0	73.7 ± 1.0	49.4 ± 2.2	56.8 ± 1.9	54.3 ± 1.8	50.0 ± 0.8	58.6 ± 2.2	45.5 ± 2.2	57.1
DeCoWA	77.9 ± 1.3	75.7 ± 0.9	39.4 ± 0.8	44.4 ± 0.8	62.1 ± 1.0	70.5 ± 0.4	48.5 ± 2.1	61.7 ± 1.9	57.4 ± 2.1	50.7 ± 3.0	59.5 ± 2.9	45.1 ± 2.2	57.7
OPS	77.5 ± 0.9	76.1 ± 0.9	39.1 ± 0.5	42.1 ± 0.4	64.4 ± 1.7	73.2 ± 0.7	48.7 ± 2.0	57.9 ± 0.9	55.0 ± 2.6	48.3 ± 1.6	58.9 ± 2.3	49.6 ± 2.1	57.6
SID	76.8 ± 0.7	75.0 ± 0.9	37.5 ± 1.0	37.5 ± 1.0	58.7 ± 2.0	68.2 ± 0.7	45.0 ± 0.9	54.6 ± 2.2	51.3 ± 1.6	46.2 ± 2.0	54.3 ± 1.8	44.3 ± 2.1	53.7
EDA	77.2 ± 1.1	74.9 ± 1.2	32.6 ± 0.5	32.7 ± 0.5	54.7 ± 2.0	65.9 ± 1.3	34.6 ± 0.4	49.5 ± 1.7	47.7 ± 2.6	42.0 ± 1.6	53.8 ± 2.4	43.1 ± 2.0	50.6

^a For each marked Qwen/Kimi column, one provider-blocked image was excluded from every input condition and all five evaluations for that model. The fixed model-specific valid set therefore contains 99 images; all other columns use 100 images.

recognition levels differ across VLM families, the within-model clean-to-adversarial change is more informative than comparing raw scores between different VLMs. The broad reduction is consistent with EDA's design goal of reducing dependence on one exact CNN input configuration by optimizing over varied spatial and appearance views. However, the VLM experiment establishes a cross-model empirical effect under the stated label-conditioned protocol; it does not isolate a VLM-specific causal mechanism, and the reported standard deviations should not be interpreted as homogeneous same-temperature repeatability across all interfaces.

5. Conclusion

This study investigated how the spatial support used during gradient estimation affects cross-architecture adversarial transfer from CNN surrogates. EDA combines two forms of controlled diversity: sampling either a movable-boundary full grid or an interior center grid changes the control-point geometry, while scale-dependent reflection expansion supports TPS resampling near the image boundary without modifying the final adversarial image. Appearance augmentation supplies additional photometric diversity but is not the source of the spatial design. The resulting take-home message is that transfer gradients become more effective on ViT targets when they are estimated over smooth deformations that vary both boundary and interior structure rather than over one fixed sampling geometry.

The source-excluded results support this conclusion across four CNN sources, three ImageNet-style datasets, four perturbation budgets, and five random seeds. The matched layout ablations consistently favor movable full-grid boundary points within both the full-grid-only and dual-layout configurations, and the padding ablation favors reflection support in the evaluated RN-18 setting. EDA also remains compatible with existing gradient-based attacks and reduces label-conditioned recognition across the evaluated VLMs. The evidence is nevertheless bounded: targeted ASR remains far below untargeted ASR, transfer to modern robust classifiers is substantially weaker than transfer to normally trained models, and the two ViT-source evaluations do not show uniform dominance.

The present analysis establishes empirical associations rather than a causal theory linking transformation support, gradient alignment, and transferability. The feature-space and Grad-CAM measurements in one RN-18-to-Swin-T setting display larger representation and attention changes under EDA, but they do not prove that those changes cause the ASR gains; likewise, the padding formula bounds sampled control-point displacement rather than every interpolated TPS location. These boundaries leave open how deformation geometry should adapt to different surrogate-target pairs and robustness regimes. Overall, EDA provides a practical design principle for transfer-based attacks: diversify the geometric support of gradient estimation while preserving the final adversarial example in the original coordinate system and within the prescribed ℓ_∞ budget.

CRedit authorship contribution statement

Jiachen Wang: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization; **Hegui Zhu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization; **Yifan Zhang:** Visualization, Validation, Investigation; **Chengqing Li:** Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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